

(12) **United States Patent**
Cho et al.

(10) **Patent No.:** **US 12,401,003 B2**
(45) **Date of Patent:** **Aug. 26, 2025**

(54) **MANUFACTURING METHOD FOR FORMING AN INORGANIC LIGHT EMITTING ELEMENT HAVING A LOW SURFACE ROUGHNESS**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/483,444**

(22) Filed: **Oct. 9, 2023**

(65) **Prior Publication Data**

US 2024/0038738 A1 Feb. 1, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/250,633, filed as application No. PCT/KR2019/000105 on Jan. 3, 2019, now Pat. No. 11,810,905.

(30) **Foreign Application Priority Data**

Aug. 16, 2018 (KR) 10-2018-0095709

(51) **Int. Cl.**

H01L 25/075 (2006.01)

H10D 86/40 (2025.01)

(Continued)

(52) **U.S. Cl.**

CPC **H01L 25/0753** (2013.01); **H10D 86/40** (2025.01); **H10D 86/60** (2025.01);
(Continued)

(58) **Field of Classification Search**

CPC H10D 86/40; H10D 86/60; H01L 25/0753;
H10K 2102/341; H10H 20/80; H10H 20/84; H10H 29/80

See application file for complete search history.

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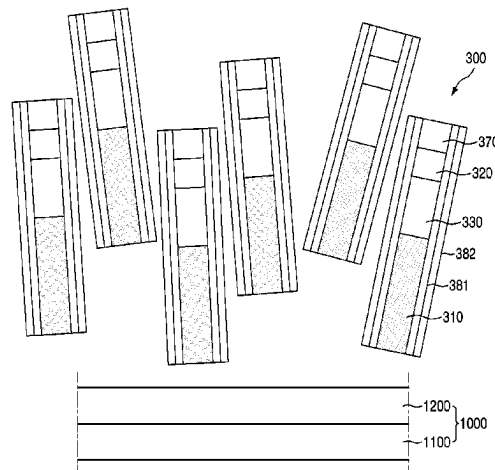
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(57)

ABSTRACT

Provided are a light emitting device, a method for manufacturing same, and a display device including the light emitting device. The method for manufacturing the light emitting device comprises the steps of: preparing a lower substrate including a substrate and a buffer semiconductor layer formed on the substrate, forming an element rod by forming a separating layer disposed on the lower substrate, forming a first conductivity type semiconductor layer, an

(Continued)



active material layer, and a second conductivity type semiconductor layer on the separating layer, and etching the first conductivity type semiconductor layer, the active material layer, the second conductivity type semiconductor layer, and the separating layer in a direction perpendicular to the lower substrate, forming a first insulating layer surrounding an outer circumferential surface of the element rod, forming a second insulating layer surrounding an outer circumferential surface of the first insulating layer and separating the element rod from the lower substrate to form a light emitting element.

11 Claims, 21 Drawing Sheets

- (51) **Int. Cl.**
H10D 86/60 (2025.01)
H10H 20/01 (2025.01)
H10H 20/82 (2025.01)
H10H 20/857 (2025.01)
- (52) **U.S. Cl.**
 CPC **H10H 20/018** (2025.01); **H10H 20/82** (2025.01); **H10H 20/857** (2025.01); **H10H 20/0364** (2025.01)

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Fig. 1

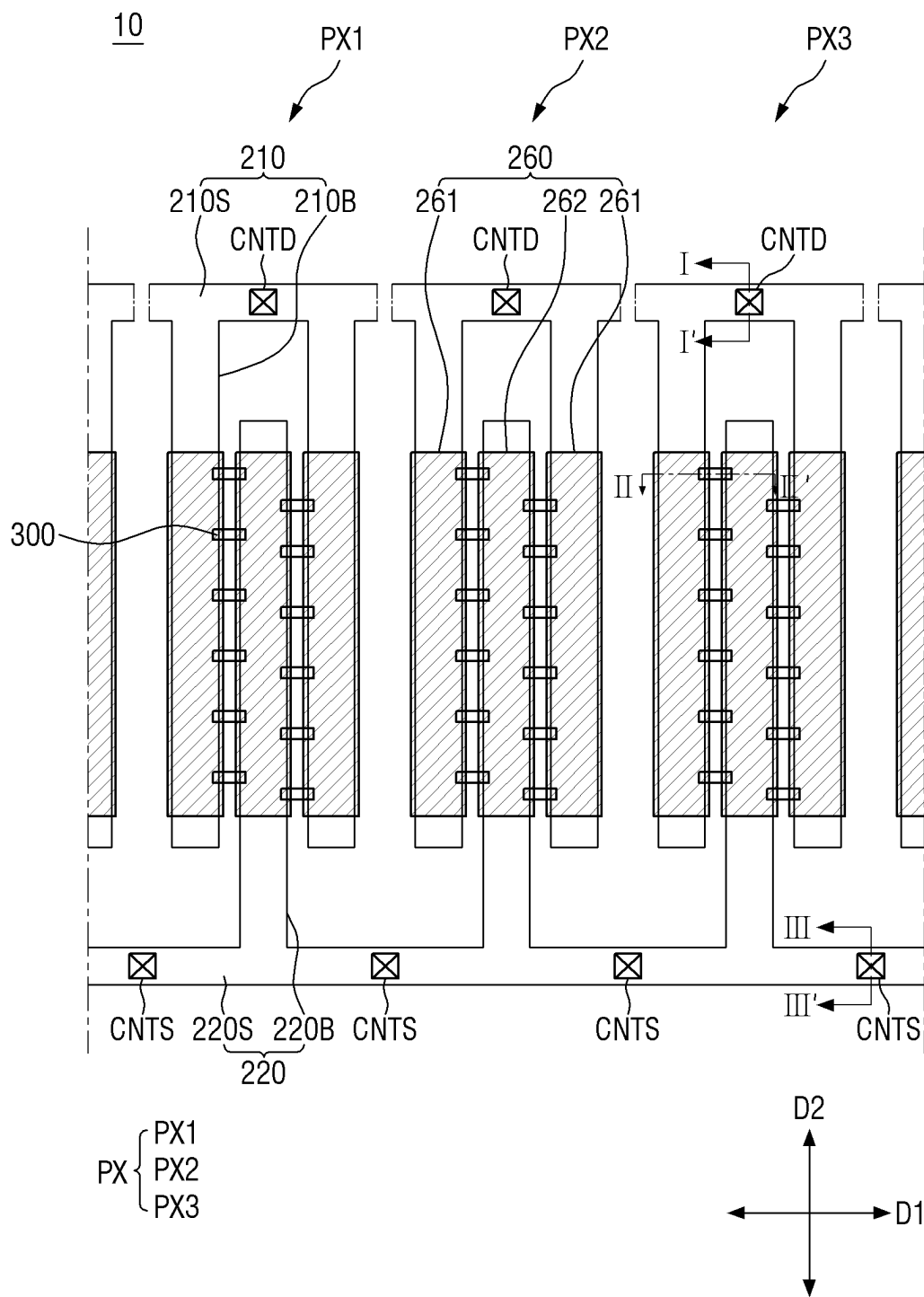


Fig. 2

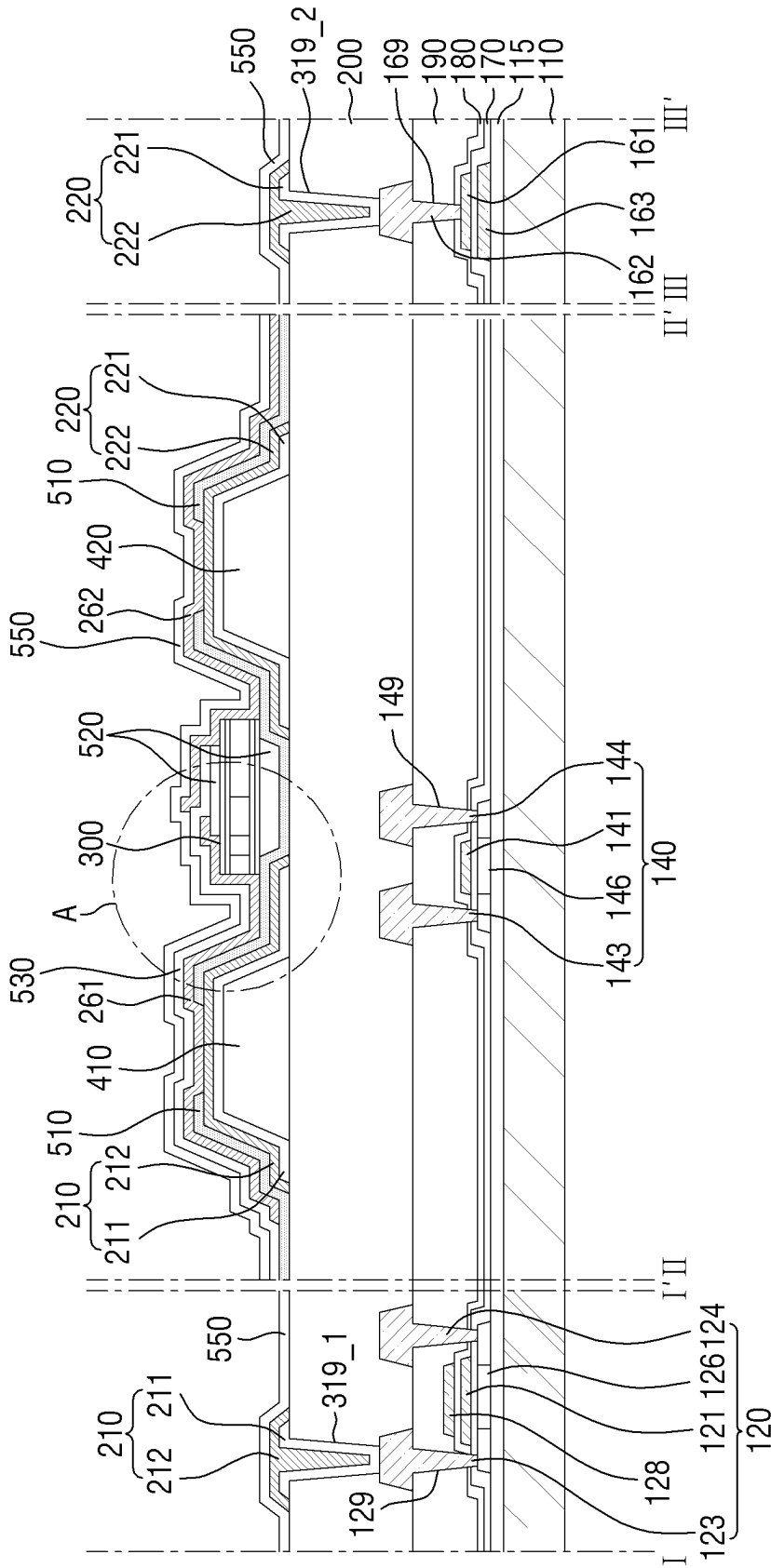


Fig. 3

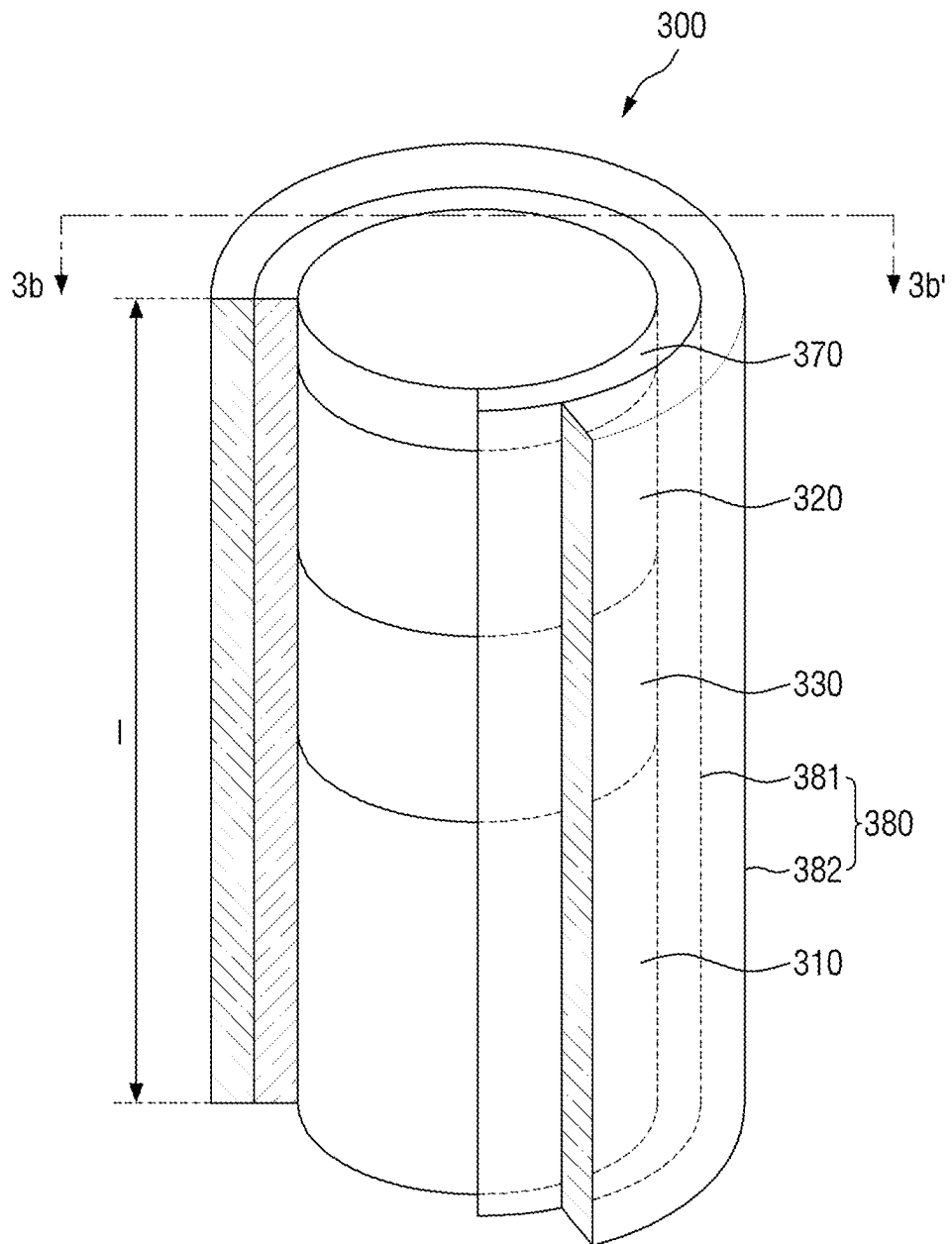
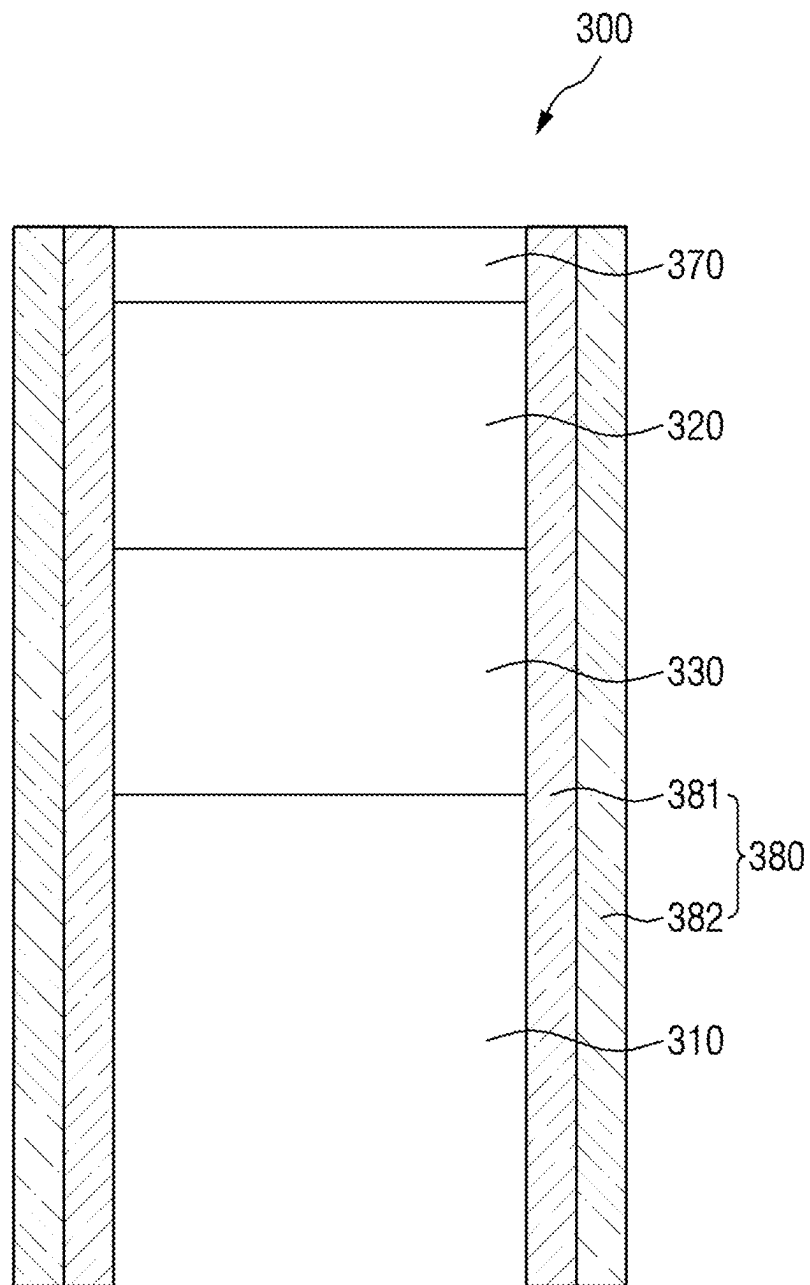


Fig. 4



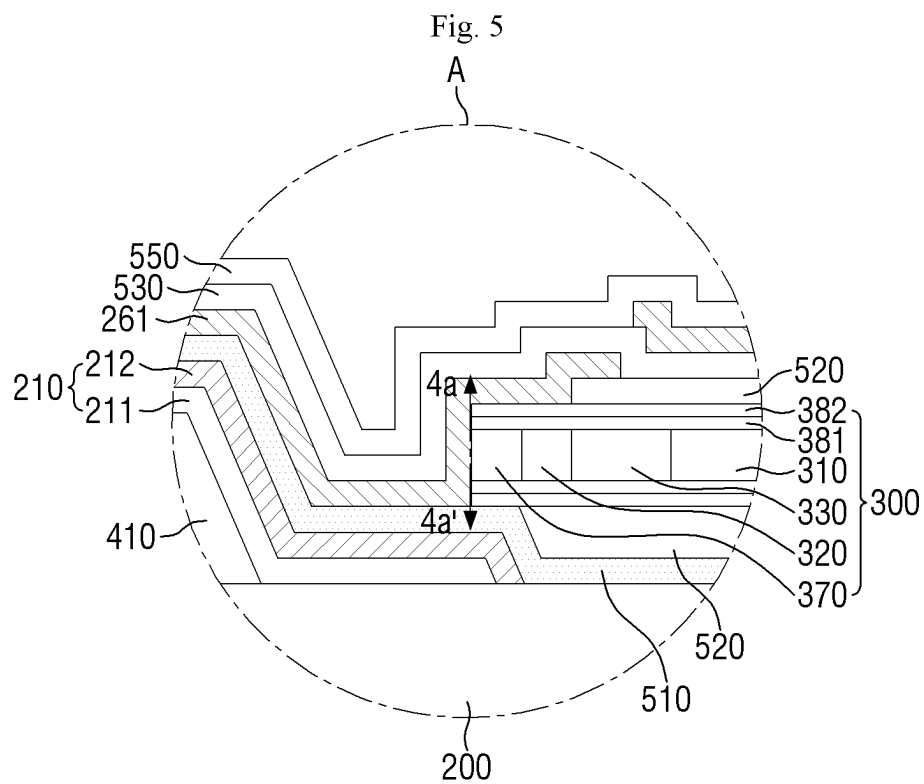


Fig. 6

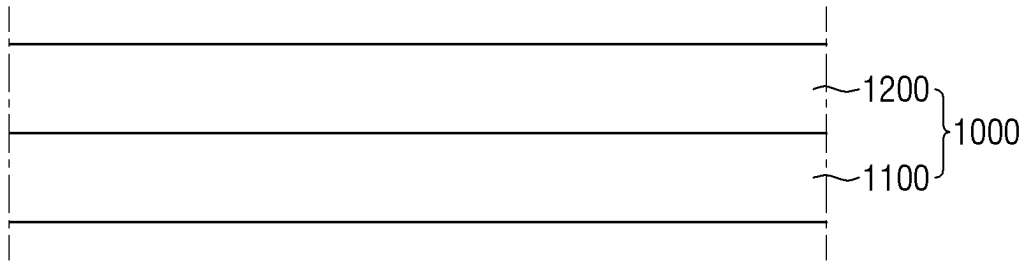


Fig. 7

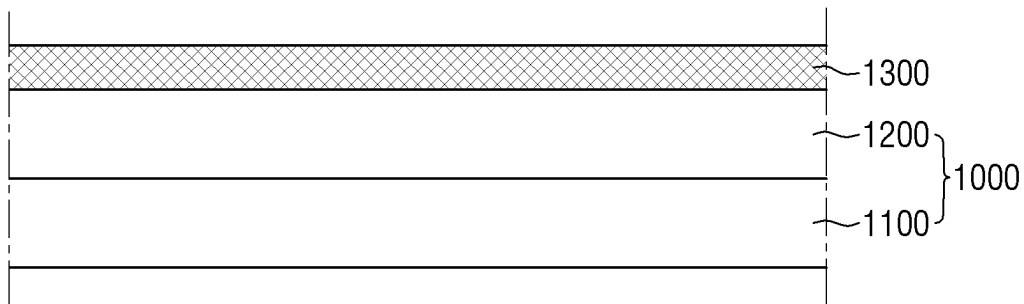


Fig. 8

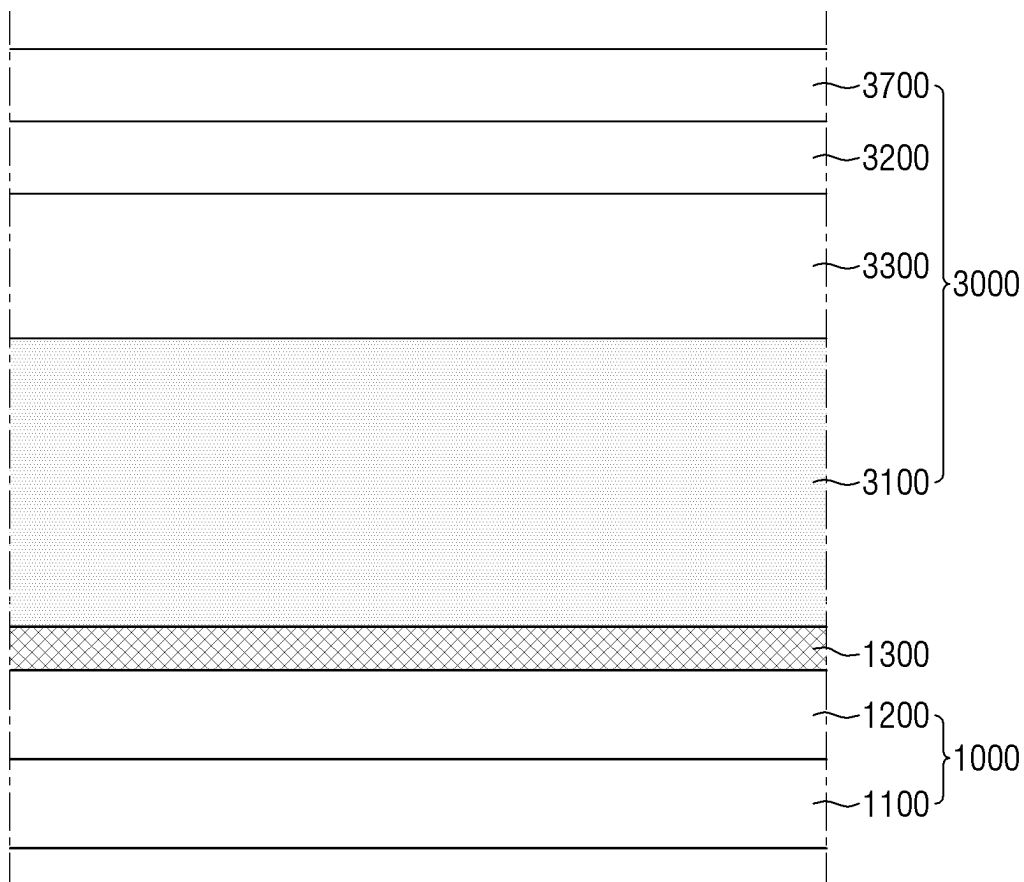


Fig. 9

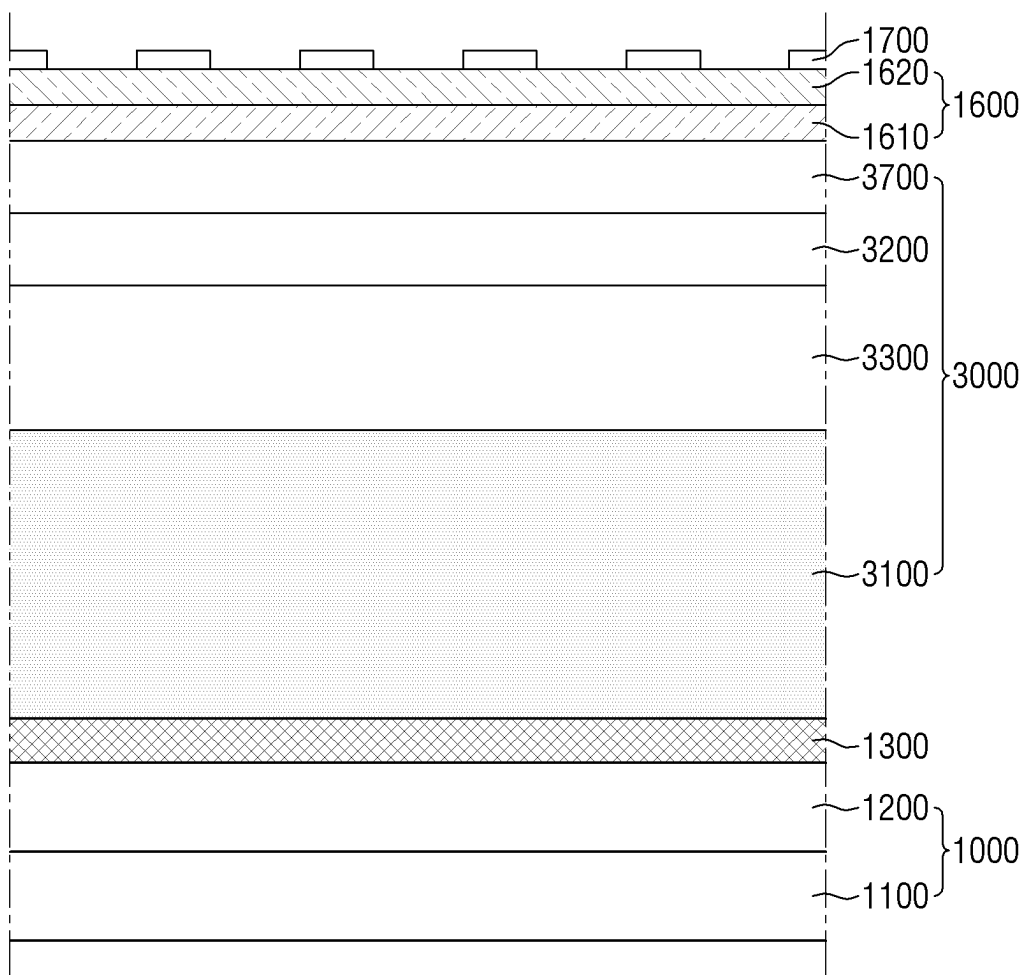


Fig. 10

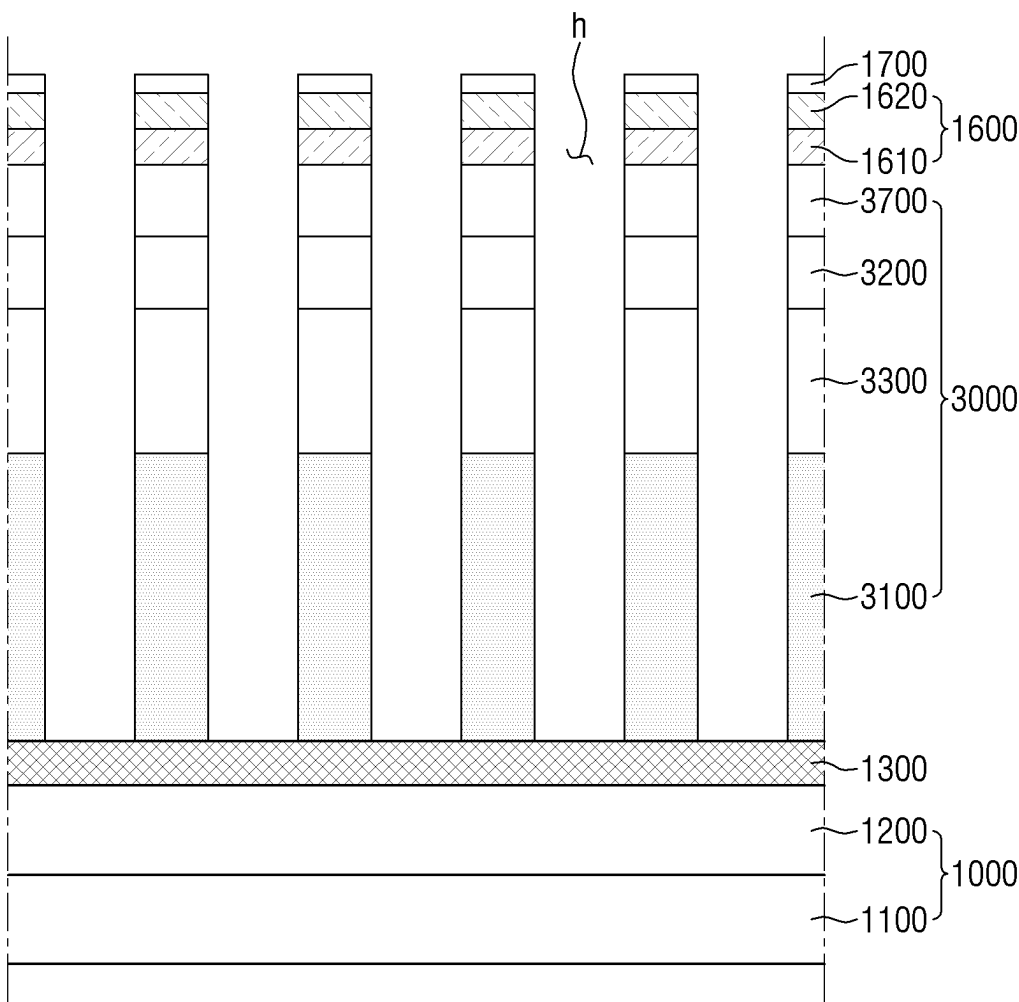


Fig. 11

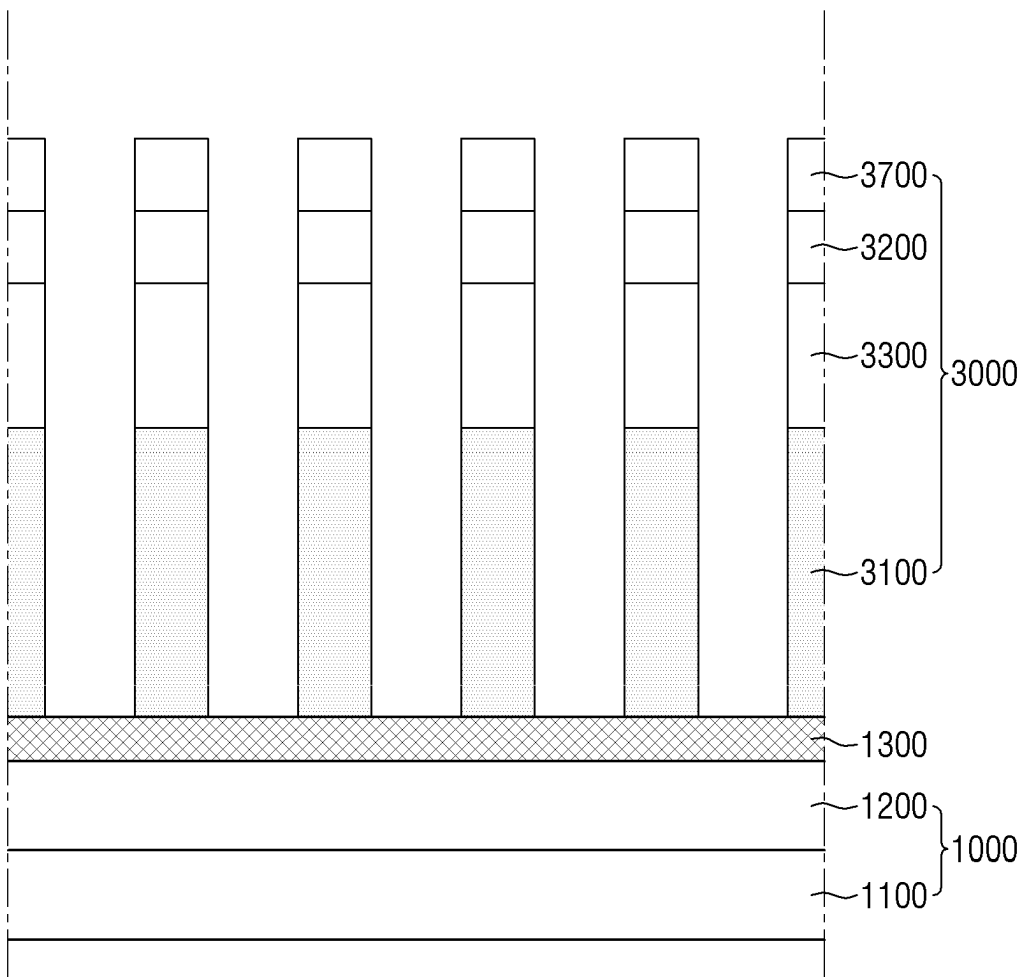


Fig. 12

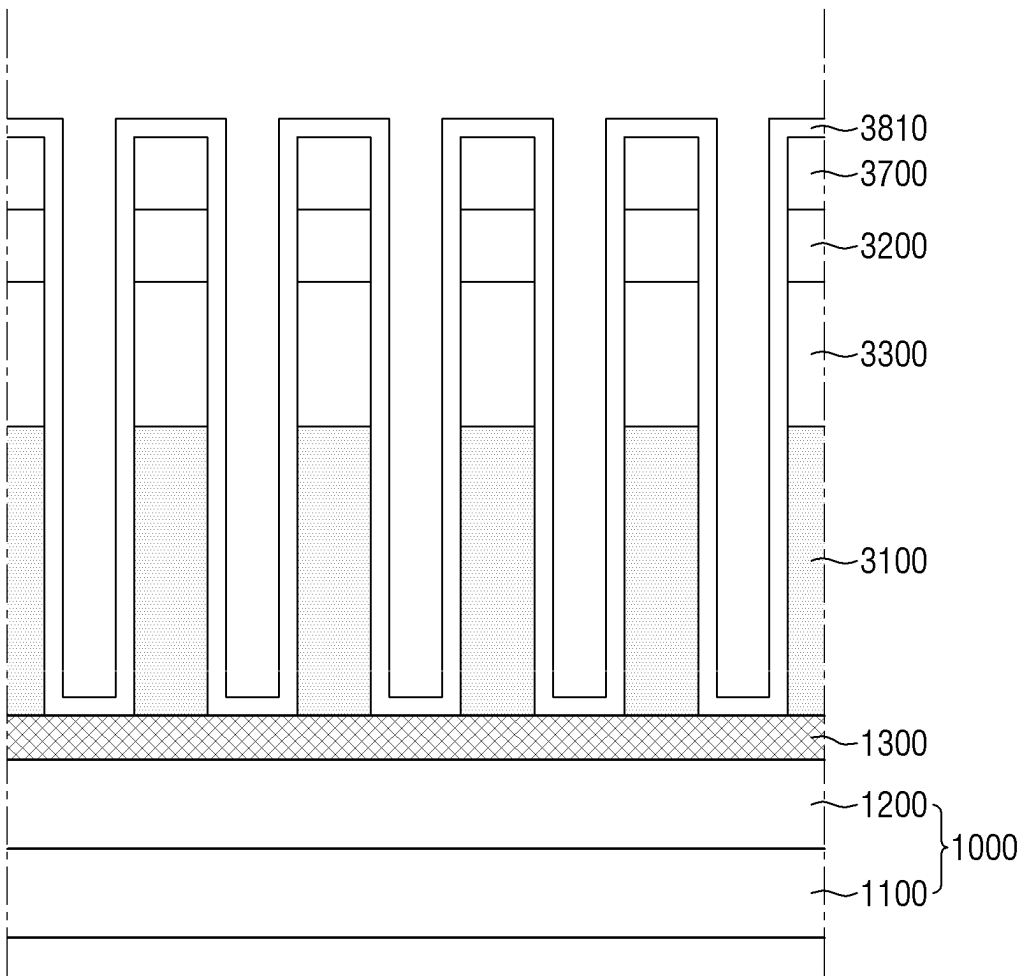


Fig. 13

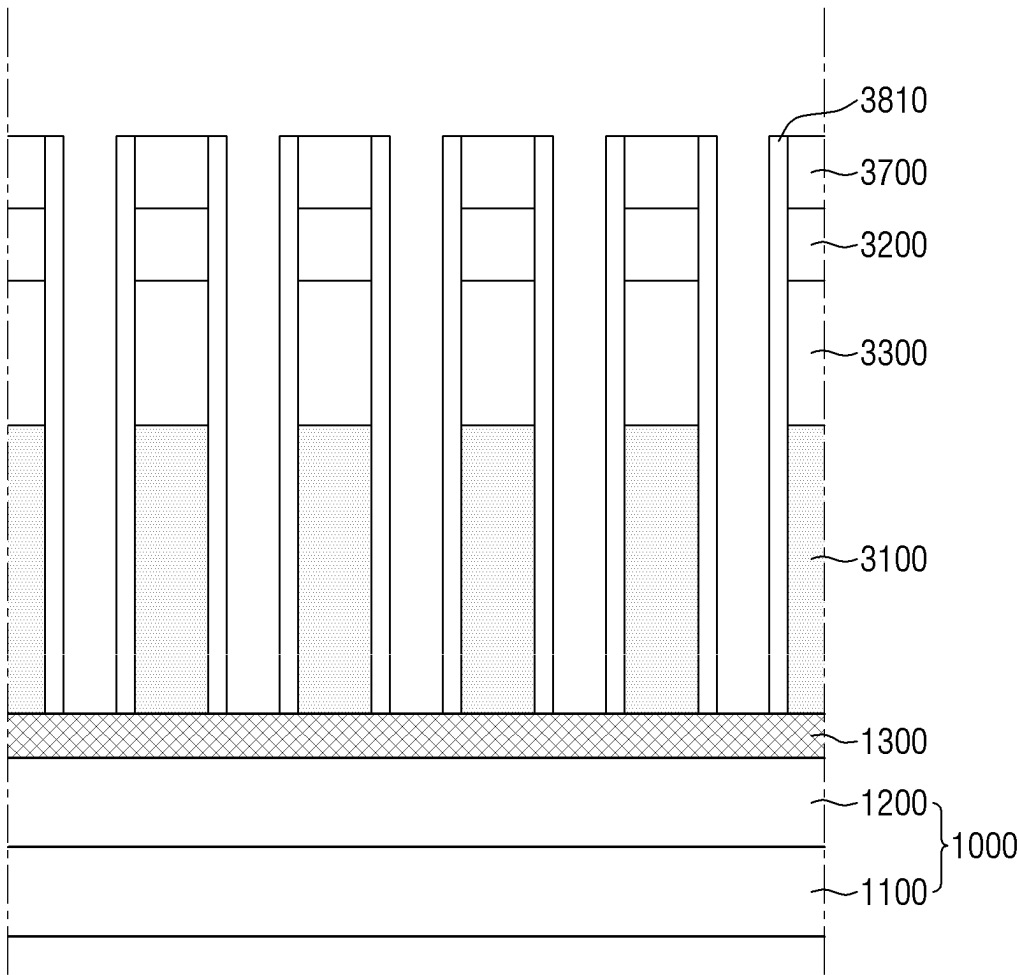


Fig. 14

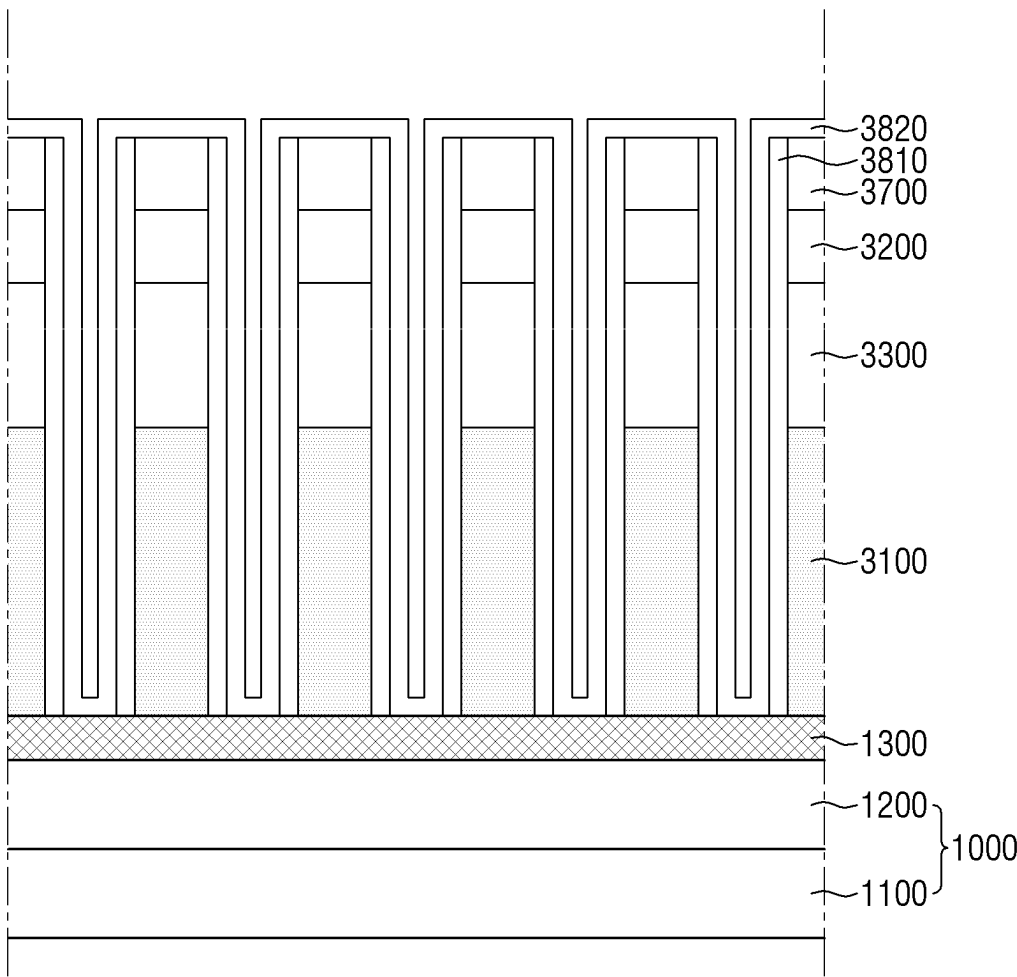


Fig. 15

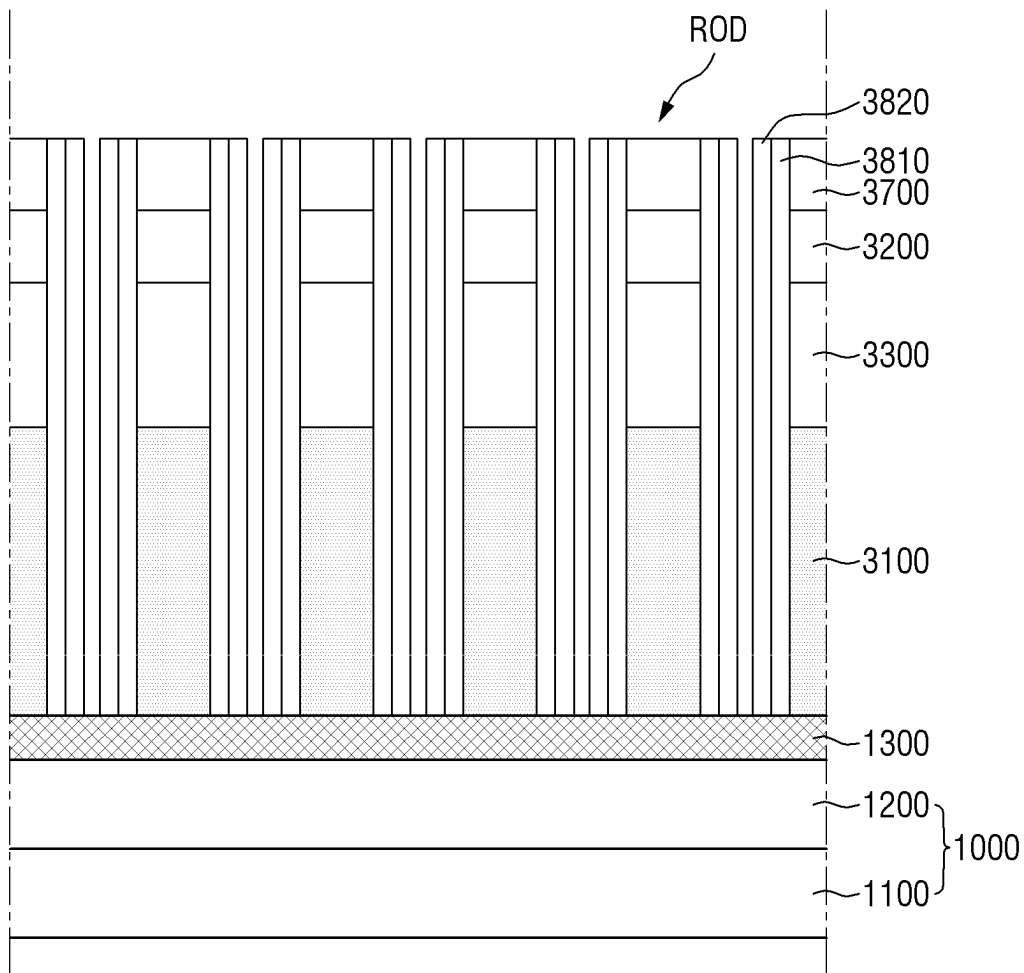


Fig. 16

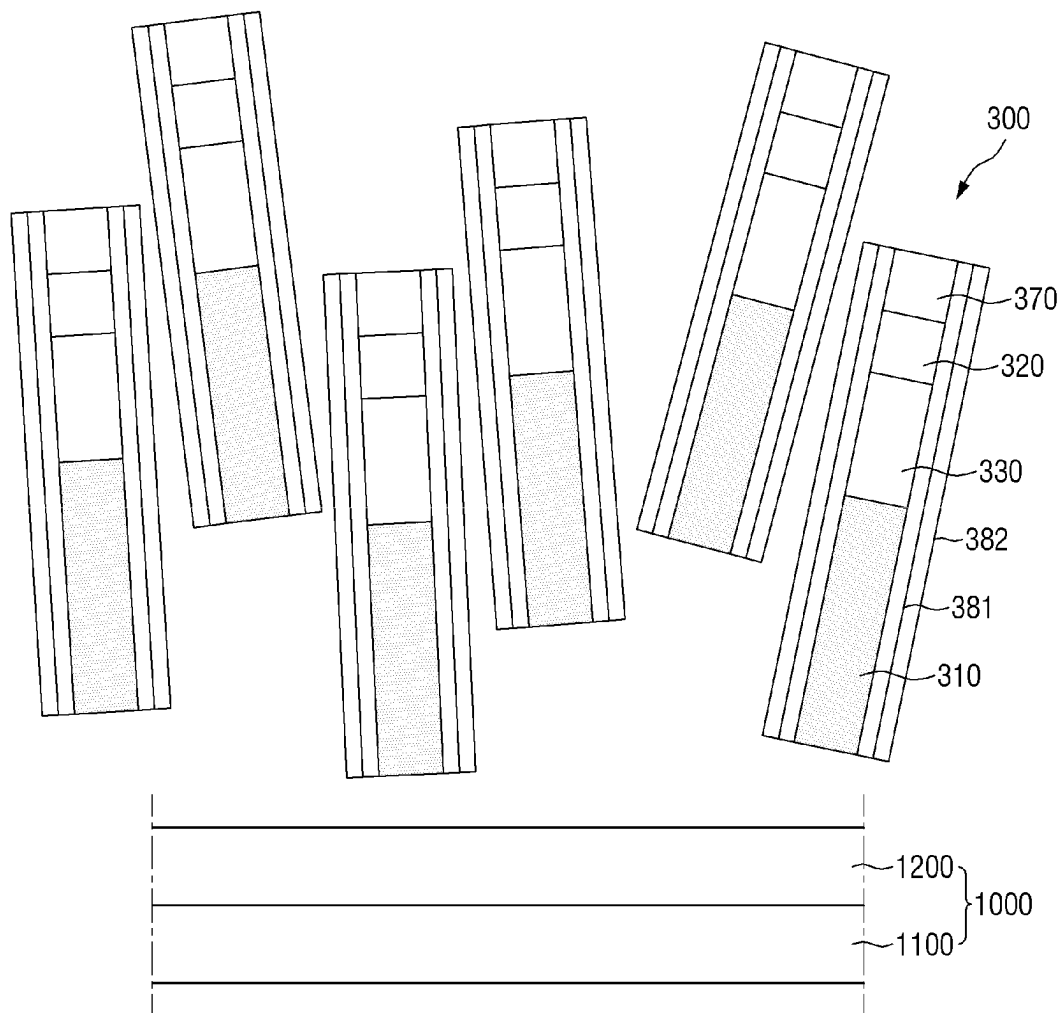


Fig. 17

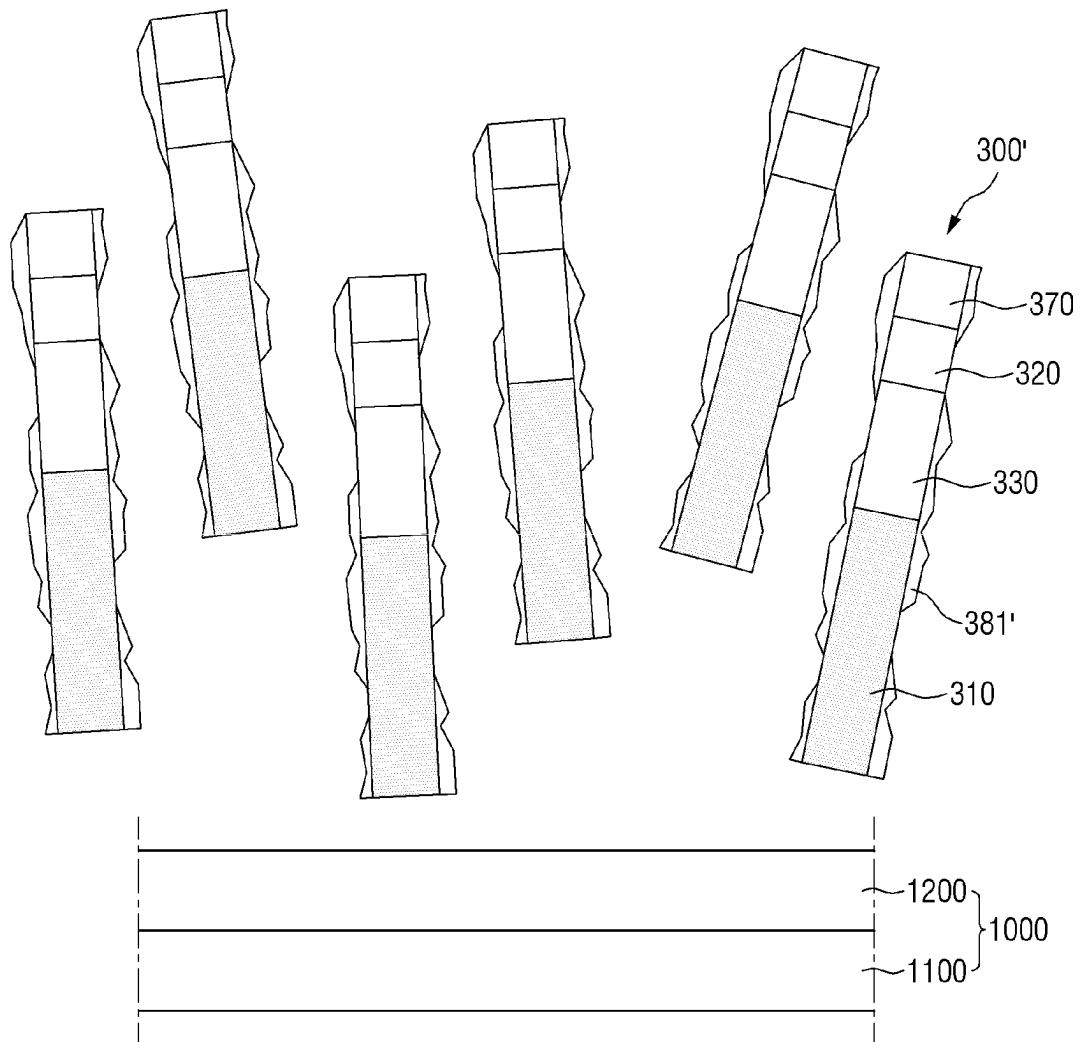


Fig. 18

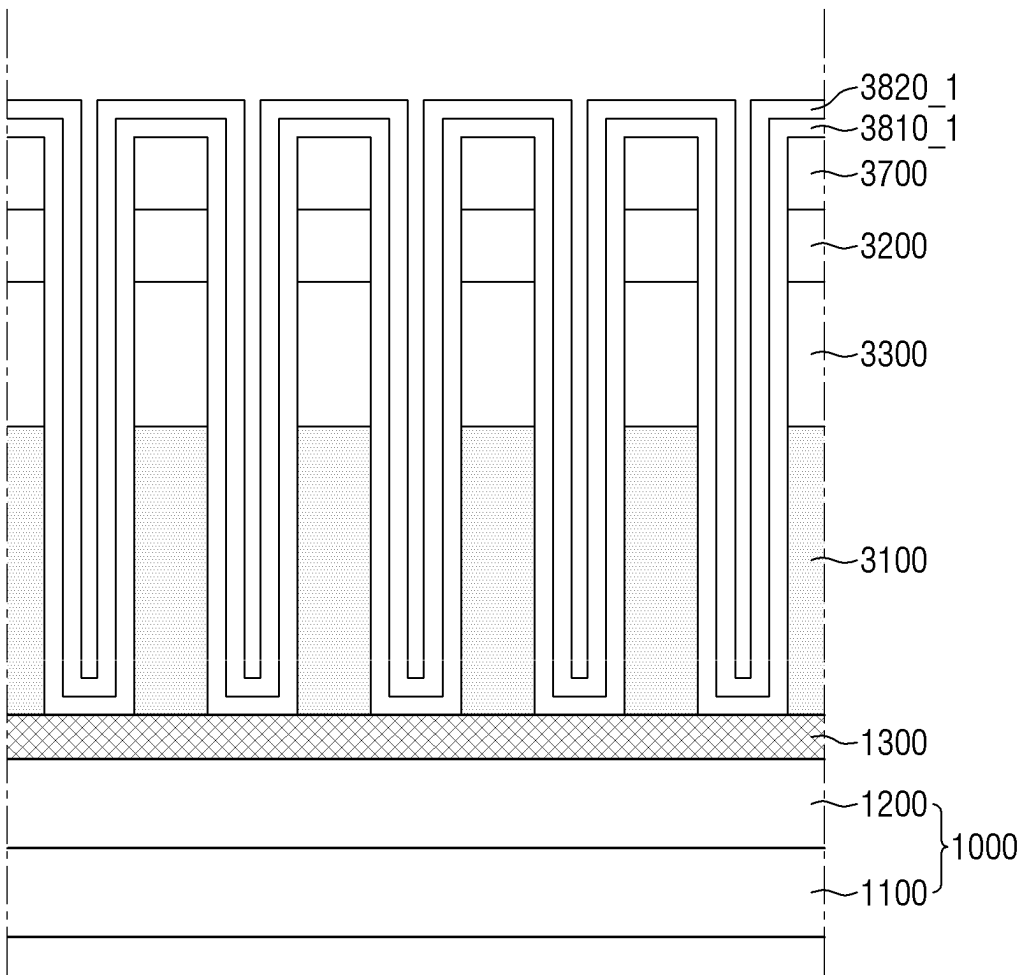


Fig. 19

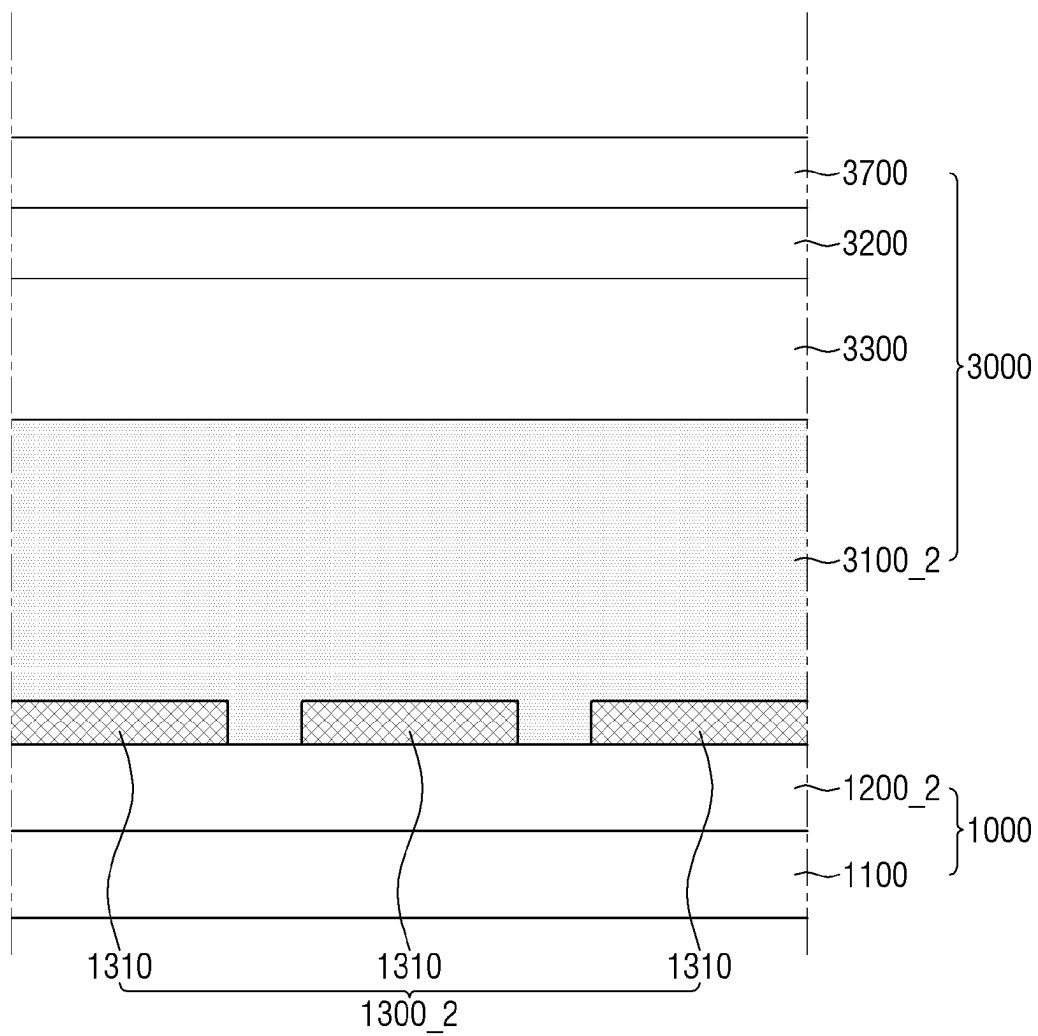


Fig. 20

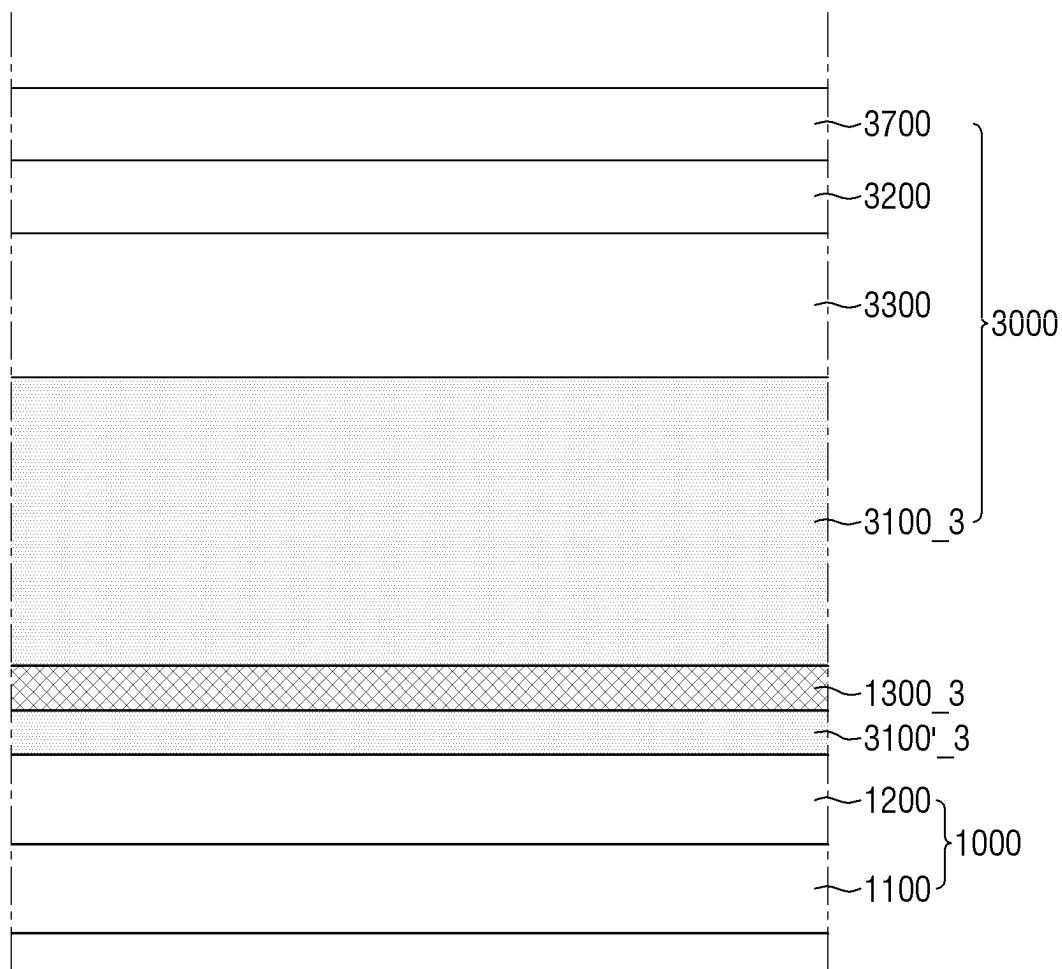
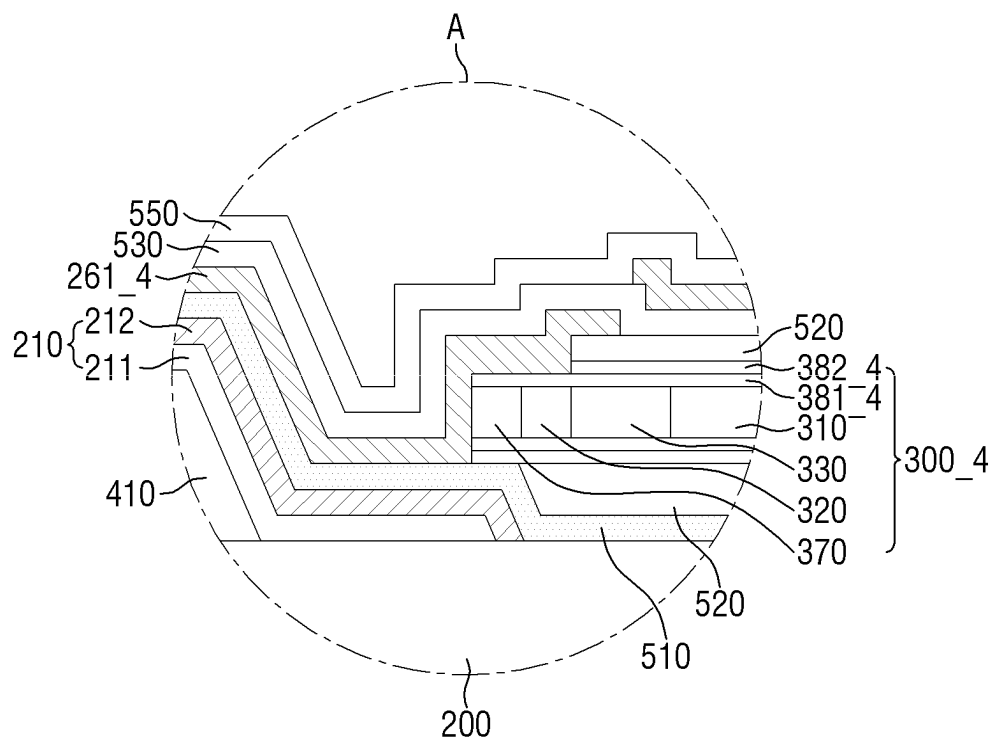


Fig. 21



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MANUFACTURING METHOD FOR FORMING AN INORGANIC LIGHT EMITTING ELEMENT HAVING A LOW SURFACE ROUGHNESS

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/250,633, filed Feb. 12, 2021, which is a U.S. National Phase Patent Application of Korean International Application No. PCT/KR2019/000105, which claims priority to Korean Patent Application No. 10-2018-0095709 filed on Aug. 16, 2018, the entire content of all of which is incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a light emitting element, a manufacturing method thereof, and a display device including the light emitting element and, in particular, to a light emitting element including a protective layer on an outer surface thereof, a manufacturing method thereof, and a display device including the light emitting element.

BACKGROUND ART

The importance of display devices has steadily increased with the development of multimedia technology. Accordingly, various types of display devices such as an organic light emitting display (OLED), a liquid crystal display (LCD) and the like have been used.

A display device is a device for displaying an image, and includes a display panel, such as an organic light emitting display panel or a liquid crystal display panel. Among them, a light emitting display panel may include a light emitting element. Examples of a light emitting diode (LED) include an organic light emitting diode (OLED) using an organic material as a fluorescent material, and an inorganic light emitting diode using an inorganic material as a fluorescent material.

The organic light emitting diode (OLED) using an organic material as a fluorescent material of a light emitting element has advantages in that a manufacturing process is simple and a display device can have flexibility. However, it is known that the organic material is vulnerable to a high-temperature operating environment and the blue light efficiency is relatively low.

DISCLOSURE

Technical Problem

The inorganic light emitting diode may be manufactured by growing an n-type or p-type doped semiconductor layer and an inorganic fluorescent material layer on a substrate, forming a rod having a specific shape, and separating the rod. However, when using a chemical method to separate the light emitting element, there is a problem that an insulating material layer surrounding the outer surface of the light emitting element is partially damaged.

In view of the above, aspects of the present disclosure provide a light emitting element including a protective layer protecting an insulating material layer on an outer circumferential surface of the light emitting element and a manufacturing method thereof.

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It should be noted that aspects of the present disclosure are not limited to the above-mentioned aspects, and other unmentioned aspects of the present disclosure will be clearly understood by those skilled in the art from the following descriptions.

Technical Solution

According to an exemplary of the present disclosure, a manufacturing method of a light emitting element, comprises preparing a lower substrate including a substrate and a buffer semiconductor layer formed on the substrate, forming an element rod by forming a separating layer disposed on the lower substrate, forming a first conductivity type semiconductor layer, an active material layer, and a second conductivity type semiconductor layer on the separating layer, and etching the first conductivity type semiconductor layer, the active material layer, the second conductivity type semiconductor layer, and the separating layer in a direction perpendicular to the lower substrate, forming a first insulating layer surrounding an outer circumferential surface of the element rod, forming a second insulating layer surrounding an outer circumferential surface of the first insulating layer and separating the element rod from the lower substrate to form a light emitting element.

The forming of the light emitting element may comprise etching and removing the separating layer by an etchant for separation containing fluorine (F), and the second insulating layer may have an etch selectivity with respect to the etchant, which is greater than an etch selectivity of the separating layer with respect to the etchant.

The second insulating layer may have an etch selectivity with respect to the etchant, which is greater than an etch selectivity of the first insulating layer with respect to the etchant.

The first insulating layer may include at least one of silicon oxide (SiO_x), aluminum oxide (Al_2O_3), or silicon oxynitride (SiO_xN_y), and the second insulating layer includes at least one of silicon nitride (SiNx), aluminum nitride (AlN), or silicon oxynitride (SiO_xN_y).

In the light emitting element, a parting surface where the element rod is separated by removing the separating layer, may be substantially flat and parallel to a top surface of the second conductivity type semiconductor layer.

In the light emitting element, the parting surface may have a surface roughness in a range of 8 nm Ra to 12 nm Ra.

The first insulating layer may have a substantially constant thickness in a long axis direction crossing both ends of the light emitting element.

The forming of the first insulating layer may comprise forming a first insulating layer disposed to cover an outer surface of the element rod and a first etching step of exposing a top surface of the element rod by etching the first insulating layer, and wherein the forming of the second insulating layer may comprise forming a second insulating layer disposed to cover the outer surface of the element rod and a second etching step of exposing the top surface of the element rod by etching the second insulating layer.

In the first etching step and the second etching step, at least a part of the separating layer in an area overlapping a separation region of the element rod may be exposed.

The second insulating layer may be formed to surround an outer surface of the first insulating layer after forming the first insulating layer, and the first etching step and the second etching step may be simultaneously performed after forming the second insulating layer.

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The forming of the element rod may further comprise forming an electrode material layer on the second conductivity type semiconductor layer.

The forming of the element rod may comprise forming an etching mask layer on the electrode material layer and an etching pattern layer having one or more nanopatterns separated from each other on the etching mask layer, forming a hole by etching an area formed by the nanopatterns being separated from each other in a direction perpendicular to the lower substrate and removing the etching mask layer and the etching pattern layer.

The first conductivity type semiconductor layer, the active material layer, the second conductivity type semiconductor layer, and the electrode material layer may include a material different in etch selectivity from the separating layer, and the forming of the hole may further comprise etching the first conductivity type semiconductor layer, the active material layer, the second conductivity type semiconductor layer, and the electrode material layer in a direction perpendicular to the lower substrate to expose at least a portion of an overlapping area between the separating layer and the area formed by the nanopatterns being separated from each other; and etching and patterning the exposed area of the separating layer.

According to another exemplary embodiment of the present disclosure, a light emitting element comprises a first conductivity type semiconductor doped with a first polarity, an active layer disposed on the first conductivity type semiconductor, a second conductivity type semiconductor formed on the active layer and doped with a second polarity opposite to the first polarity, an electrode material layer disposed on the second conductivity type semiconductor and a first insulating layer surrounding side surfaces of the first conductivity type semiconductor, the second conductivity type semiconductor, the active layer and the electrode material layer, and a second insulating layer surrounding an outer circumferential surface of the first insulating layer, wherein the first insulating layer and the second insulating layer are different in etch selectivity.

An etch selectivity of the second insulating layer with respect to the etchant containing fluorine (F) may be greater than an etch selectivity of the first insulating layer with respect to the etchant.

The first insulating layer includes at least one of silicon oxide (SiOx), aluminum oxide (Al₂O₃), or silicon oxynitride (SiOxNy), and the second insulating layer includes at least one of silicon nitride (SiNx), aluminum nitride (AlN), or silicon oxynitride (SiOxNy).

A bottom surface of the first conductivity type semiconductor and a top surface of the second conductivity type semiconductor may be substantially flat and parallel to each other, and the bottom surface of the first conductivity type semiconductor and the top surface of the second conductivity type semiconductor may have a surface roughness in a range of 8 nm Ra to 12 nm Ra.

According to the other exemplary of the present disclosure, a display device comprises a substrate, at least one first electrode and at least one second electrode extending in a first direction on the substrate and spaced apart from each other in a second direction different from the first direction, at least one light emitting element disposed in a separation space between the first electrode and the second electrode, a first contact electrode partially covering the first electrode and contacting a first end of the light emitting element, and a second contact electrode spaced apart from the first contact electrode and partially covering the second electrode to contact a second end opposite to the first end of the light

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emitting element, wherein the light emitting element includes an element rod, a first insulating layer surrounding an outer circumferential surface of the element rod, and a second insulating layer surrounding at least a portion of an outer circumferential surface of the first insulating layer.

The element rod may include a first conductivity type semiconductor doped with a first polarity, an active layer disposed on the first conductivity type semiconductor, a second conductivity type semiconductor formed on the active layer and doped with a second polarity opposite to the first polarity, and an electrode material layer formed on the second conductivity type semiconductor, wherein the first insulating layer may surround side surfaces of the first conductivity type semiconductor, the active layer, the second conductivity type semiconductor, and the electrode material layer, and the second insulating layer may include a material different in etch selectivity from the insulating material layer.

In the second insulating layer, top surfaces of the first end and the second end of the light emitting element in cross-sectional view may be patterned to partially expose the first insulating layer, and the first contact electrode and the second contact electrode may be partially in contact with the exposed first insulating layer.

The above and other features and advantages of the present invention will become more apparent by describing in detail exemplary embodiments thereof with reference to the attached drawings.

Advantageous Effects

According to a manufacturing method of a light emitting element according to an embodiment, it is possible to manufacture a light emitting element in which an insulating material layer and an insulating protective layer, different in etch selectivity, are disposed on an outer circumferential surface of the light emitting element. Although the light emitting element is formed by a chemical lift off (CLO) method during manufacture of the light emitting element, the insulating protective layer can protect the insulating material layer to have a constant thickness without being damaged by an etchant for separation.

The light emitting element being arranged between two electrodes of a display device has two end surfaces that are flat and substantially parallel, which is capable of preventing an open or short circuit problem of a contact electrode material from occurring in the case of connection with a contact electrode.

Advantageous effects according to the present disclosure are not limited to those mentioned above, and various other advantageous effects are included herein.

DESCRIPTION OF DRAWINGS

FIG. 1 is a plan view of a display device according to an embodiment;

FIG. 2 is a cross-sectional view taken along lines I-I', II-II' and III'-III' of FIG. 1;

FIG. 3 is a schematic view of a light emitting element according to an embodiment;

FIG. 4 is a cross-sectional view taken along line 3b-3b' of FIG. 3A;

FIG. 5 is an enlarged view of part A of FIG. 2;

FIGS. 6 to 16 are schematic cross-sectional views schematically showing a method for manufacturing a light emitting element according to an embodiment;

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FIG. 17 is a schematic view showing a part of a method of manufacturing a light emitting element according to a comparative example;

FIG. 18 is a schematic view showing a part of a method of manufacturing a light emitting element according to another embodiment;

FIGS. 19 and 20 are cross-sectional views schematically showing the arrangement of a separating layer in a semiconductor structure according to another embodiment; and

FIG. 21 is a cross-sectional view illustrating a portion of a display device according to another embodiment.

MODES OF THE INVENTION

The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.

It will also be understood that when a layer is referred to as being “on” another layer or substrate, it can be directly on the other layer or substrate, or intervening layers may also be present. The same reference numbers indicate the same components throughout the specification.

It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. For instance, a first element discussed below could be termed a second element without departing from the teachings of the present invention. Similarly, the second element could also be termed the first element.

Hereinafter, exemplary embodiments will be described with reference to the accompanying drawings.

FIG. 1 is a plan view of a display device according to an embodiment.

The display device 10 may include at least one area defined as a pixel PX. The display device 10 may include a display area composed of a plurality of pixels PX, each emitting light in a specific wavelength band to the outside of the display device 10. Although three pixels PX1, PX2, and PX3 are exemplarily illustrated in FIG. 1, it is obvious that the display device 10 may include a larger number of pixels. Although it is shown in the drawing that a plurality of pixels PX are arranged in one direction, e.g., first direction D1, in cross-sectional view, the plurality of pixels PX may also be arranged in the second direction D2 crossing the first direction D1. Further, each of the pixels PX of FIG. 1 may be divided into a plurality of portions, and each portion may constitute one pixel PX. The pixels are not necessarily arranged in parallel only in the first direction D1 as shown in FIG. 1, and may have various structures, such as being arranged in a vertical direction (or the second direction D2) or in a zigzag manner.

Although not shown in the drawing, the display device 10 may include an emission area in which light emitting elements 300 are arranged for emitting certain color lights, and a non-emission area defined as an area remaining after exclusion of the emission area. The non-emission area may be covered by certain members that are not visually perceived from the outside of the display device 10. Various members for driving the light emitting elements 300 disposed in the emission area may be disposed in the non-

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emission area. For example, the non-emission area may include a wiring, a circuit unit, and a driving unit for applying an electrical signal to the emission area, but the present disclosure is not limited thereto.

The plurality of pixels PX may display colors by including one or more light emitting elements 300 emitting light of a specific wavelength band. The light emitted from the light emitting element 300 may be projected to the outside through the emission area of the display device 10. In an embodiment, each of the pixels PX presenting different colors may include different light emitting elements 300 emitting different color lights. For example, a first pixel PX1 presenting a red color may include a light emitting element 300 emitting a red light, a second pixel PX2 presenting a green color may include a light emitting element 300 emitting a green light, and a third pixel PX3 presenting a blue color may include a light emitting element 300 emitting a blue light. However, the present disclosure is not limited thereto, and the pixels presenting different colors may, in some cases, include the light emitting elements 300 emitting the same color light (e.g., blue light), or they may each include a wavelength conversion layer or a color filter on a light emission path, to produce pixel-specific colors. However, the present disclosure is not limited thereto, and adjacent pixels PX may emit the same color light in some cases.

With reference to FIG. 1, the display device 10 may include a plurality of electrodes 210 and 220 and a plurality of light emitting elements 300. At least a portion of each of the electrodes 210 and 220 may be arranged in each pixel PX, and electrically connected to the light emitting elements 300 to apply an electrical signal, in order for the light emitting elements 300 to emit a certain color light.

At least a portion of each of the electrodes 210 and 220 may also contribute to producing an electric field in the pixels PX, to align the light emitting elements 300. In more detail, it is necessary to precisely align the pixel-specific (PX-specific) light emitting elements 300 during the alignment of the light emitting elements 300 emitting different color lights in the plurality of pixels PX. In the case of using an electrophoresis method for aligning the light emitting elements 300, the light emitting elements 300 may be aligned in a way of depositing a solution including the light emitting elements 300 on the display device 10, and applying alternating power thereto to create a capacitance with an electric field, which produces an electrophoresis force to the light emitting elements 300.

The plurality of electrodes 210 and 220 may include a first electrode 210 and a second electrode 220. In an exemplary embodiment, the first electrode 210 may be a pixel electrode branched to each pixel PX, and the second electrode 220 may be a common electrode connected in common to the plurality of pixels PX. One of the first and second electrodes 210 and 220 may be an anode electrode of the light emitting element 300, and the other may be a cathode electrode of the light emitting element 300. However, the present disclosure is not limited thereto, and the reverse may also be the case.

The first and second electrodes 210 and 220 may include respective electrode stems 210S and 220S arranged to extend in the first direction D1 and at least one respective electrode branches 210B and 220B extending, in the second direction D2 crossing the first direction D1, from the respective electrode stems 210S and 220S.

In detail, the first electrode 210 may include a first electrode stem 210S arranged to extend in the first direction D1, and at least one first electrode branch 210B branched

from the first electrode stem **210S** and extending in the second direction **D2**. Although not shown in the drawing, the first electrode stem **210S** may be connected, at one end thereof, to a signal input pad and extend, at the other end thereof, in the first direction **D1**, maintaining electrical disconnection between the pixels **PX**. The signal input pad may be connected to a power source of the display device **10** or the outside, to apply an electrical signal or, in the case of aligning the light emitting elements **300**, alternating power to the first electrode stem **210S**.

The first electrode stem **210S** of one pixel may be arranged substantially on the same line as the first electrode stem **210S** of neighboring pixels belonging to the same row (e.g., adjacent in the first direction **D1**). That is, the first electrode stem **210S** of one pixel may be arranged such that two ends thereof terminate between corresponding pixels while being spaced apart from each other, and the first electrode stems **210S** of the neighboring pixels may be aligned with an extension line of the first electrode stem **210S** of the one pixel. In this manner, the first electrode stem **210S** may be arranged in a way of being formed as an continuous stem electrode in a manufacturing process, and cut off by a laser or the like to be open after performing the alignment process of the light emitting elements **300**. Accordingly, the first electrode stems **210S** of the respective pixels **PX** may apply different electrical signals to the respective first electrode branches **210B**, which may operate independently of each other.

The first electrode branch **210B** may be branched from at least part of the first electrode stem **210S** and extend in the second direction **D2**, and may terminate to keep a distance from the second electrode stem **220S** arranged to face the first electrode stem **210S**. That is, the first electrode branch **210B** may be arranged to be connected, at one end thereof, to the first electrode stem **210S** and placed, at the other end thereof, inside the pixel **PX**, keeping a distance from the second electrode stem **220S**. The first electrode branch **210B** may be connected to the first electrode stem **210S**, which is electrically separate per pixel **PX**, so as to receive a different electrical signal per pixel **PX**.

It may also be possible that one or more first electrode branches **210B** are arranged per pixel **PX**. Although it is shown in FIG. 1 that two first electrode branches **210B** are arranged and the second electrode branch **220B** is arranged therebetween, the present disclosure is not limited thereto, and more first electrode branches **210B** may be arranged. In this case, the first electrode branches **210B** may be arranged alternately, to be separated from the plurality of second electrode branches **220B**, such that a plurality light emitting elements **300** are arranged therebetween. In some embodiments, the second electrode branch **220B** may be arranged between the first electrode branches **210B** such that each pixel **PX** is symmetrical about the second electrode branch **220B**. However, the present disclosure is not limited thereto.

The second electrode **220** may include a second electrode stem **220S** arranged to extend in the first direction **D1** and face the first electrode stem **210S**, keeping a distance from the first electrode stem **210S**, and at least one second electrode branch **220B** branched from the second electrode stem **220S** to extend in the second direction **D2** and face the first electrode branch **210B**, keeping a distance from the first electrode branch **210B**. The second electrode stem **220S** may also be connected to the signal input pad at one end thereof, like the first electrode stem **210S**. However, the second electrode stem **220S** may extend, at the other end thereof, in the first direction **D1** toward the a plurality of adjacent pixels **PX**. That is, the second electrode stem **220S**

may be electrically continuous between individual pixels **PX**. Accordingly, the second electrode stem **220S** of a certain pixel is connected at opposite ends thereof to one ends of the second electrode stems **220S** of the neighboring pixels between the pixels **PX** to apply the same electrical signal to each pixel **PX**.

The second electrode branch **220B** may be branched from at least part of the second electrode stem **220S** and extend in the second direction **D2**, and may terminate to keep a distance from the first electrode stem **210S**. That is, the second electrode branch **220B** may be arranged to be connected at one end thereof to the second electrode stem **220S**, and placed at the other end thereof inside the pixel **PX**, keeping a distance from the first electrode stem **210S**. The second electrode branch **220B** may be connected to the second electrode stem **220S**, which is electrically continuous to the respective pixels **PX**, so as to receive the same electrical signal for each pixel **PX**.

The second electrode branch **220B** may be arranged to face the first electrode branch **210B** keeping a distance from the first electrode branch **210B**. Here, the first and second electrode stems **210S** and **220S** face each other about the center of each pixel **PX**, keeping a distance, such that the first and second electrode branches **210B** and **220B** extend in the opposite directions. That is, the first electrode branch **210B** may extend to one orientation of the second direction **D2** and the second electrode branch **220B** may extend to the other orientation of the second direction **D2**, such that one ends of the respective branches are arranged to face opposite orientations about the center of the pixel **PX**. However, the present disclosure is not limited thereto, and the first and second electrode stems **210S** and **220S** may be arranged to face the same orientation about the center of the pixel **PX**, keeping a distance from each other. In this case, the first and second electrode branches **210B** and **220B** branched from the respective electrode stems **210S** and **220S** may extend in the same direction.

Although it is shown in FIG. 1 that one second electrode branch **220B** is arranged in each pixel **PX**, the present disclosure is not limited thereto, and more second electrode branches **220B** may be arranged.

The plurality of light emitting elements **300** may be aligned between the first and second electrode branches **210B** and **220B**. In detail, at least part of the plurality of the light emitting elements **300** are each electrically connected at one end thereof to the first electrode branch **210B** and at the other end thereof to the second electrode branch **220B**.

The plurality of light emitting elements **300** may be aligned substantially in parallel with one another keeping a distance in the second direction **D2**. The interval between the light emitting elements **300** is not particularly limited. One plurality of light emitting elements **300** may be adjacently arranged to form a cluster while another plurality of light emitting elements **300** may be arranged keeping a predetermined distance from one another to form a cluster, and they may also be aligned to face one orientation at a non-uniform density.

The first and second electrode branches **210B** and **220B** may have respective contact electrodes **260** arranged thereon.

The plurality of contact electrodes **260** may be arranged to extend in the second direction **D2**, and spaced apart from one another in the first direction **D1**. The contact electrodes **260** may contact at least one ends of the light emitting elements **300**, and may contact the first and second electrodes **210** and **220** to receive an electrical signal. Accordingly, the contact electrodes **260** may transfer the electrical

signal received through the first and second electrodes **220** to the light emitting elements **300**.

The contact electrodes **260** may include a first contact electrode **261** and a second contact electrode **262** arranged on the respective electrode branches **210B** and **220B** to partially cover them and contact one or the other ends of the light emitting elements **300**.

The first contact electrode **261** may be arranged on the first electrode branch **210B** to contact one ends of the light emitting elements **300** that are electrically connected to the first electrode **210**. The second contact electrode **262** may be arranged on the second electrode branch **220B** to contact the other ends of the light emitting elements **300** that are electrically connected to the second electrode **220**.

In some embodiments, the opposite ends of each of the light emitting elements **300** that are each electrically connected to the first electrode branch **210B** or the second electrode branch **220B** may be an n-type or p-type doped conductive semiconductor layer. In the case where one end of a light emitting element **300** that is electrically connected to the first electrode branch **210B** is a p-type doped conductive semiconductor layer, the other end of the light emitting element **300** that is electrically connected to the second electrode branch **220B** may be an n-type doped conductive semiconductor layer. However, the present disclosure is not limited thereto, and an opposite case may also be possible.

The first and second contact electrodes **261** and **262** may be arranged to partially cover the respective first and second electrode branches **210B** and **220B**. As shown in FIG. 1, the first and second contact electrodes **261** and **262** may be arranged to extend in the second direction **D2**, and face each other keeping a distance. However, the first and second contact electrodes **261** and **262** may terminate at one ends thereof to expose one ends of the respective electrode branches **210B** and **220B**. The first and second contact electrodes **261** and **262** may also terminate at the other ends thereof so as not to overlap the respective electrode stems **210S** and **220S** and be spaced apart therefrom. However, the present disclosure is not limited thereto, the first and second contact electrodes **261** and **262** may cover the respective electrode branches **210B** and **220B**.

Meanwhile, as shown in FIG. 1, the first and second electrode stems **210S** and **220S** may be electrically connected to a thin film transistor **120** or a power wiring **161** (to be described later) via respective contact holes, e.g., a first electrode contact hole CNTD and a second electrode contact hole CNTS. Although it is shown in FIG. 1 that the first and second electrode stems **210S** and **220S** each have a contact hole arranged thereon per pixel PX, the present disclosure is not limited thereto. Because the second electrode stem **220S** may extend to establish an electrical connection with the adjacent pixels PX as described above, the second electrode stem **220S** may, in some embodiments, be electrically connected to the thin film transistor via one contact hole.

A description is made hereinafter of the configuration of the plurality of members arranged on the display device **10** in more detail with reference to FIG. 2.

FIG. 2 is a cross-sectional view taken along lines I-I', II-II' and III'-III' of FIG. 1. Although FIG. 2 shows a single pixel PX, the configuration may be identically applicable to other pixels. FIG. 2 shows a cross section across one and the other ends of a certain light emitting element **300**.

Referring to FIGS. 1 and 2, the display device **10** may include a substrate **110**, thin film transistors **120** and **140** disposed on the substrate **110**, and the electrodes **210** and **220** disposed on the thin film transistors **120** and **140**, and

the light emitting elements **300**. The thin film transistors may include a first thin film transistor **120** and a second thin film transistor **140**, and they may be a driving transistor and a switching transistor, respectively. Each of the thin film transistors **120** and **140** may include an active layer, a gate electrode, a source electrode, and a drain electrode. The first electrode **210** may be electrically connected to the drain electrode of the first thin film transistor **120**.

Specifically, the substrate **110** may be an insulating substrate. The substrate **110** may be made of an insulating material such as glass, quartz, or polymer resin. Examples of the polymer material may include polyethersulphone (PES), polyacrylate (PA), polyarylate (PAR), polyetherimide (PEI), polyethylene naphthalate (PEN), polyethylene terephthalate (PET), polyphenylene sulfide (PPS), polyallylate, polyimide (PI), polycarbonate (PC), cellulose triacetate (CAT), cellulose acetate propionate (CAP), or a combination thereof. Further, the substrate **110** may be a rigid substrate, but may also be a flexible substrate which can be bent, folded or rolled.

A buffer layer **115** may be disposed on the substrate **110**. The buffer layer **115** can prevent diffusion of impurity ions, prevent penetration of moisture or external air, and perform a surface planarization function. The buffer layer **115** may include silicon nitride, silicon oxide, silicon oxynitride, or the like.

A semiconductor layer is disposed on the buffer layer **115**. The semiconductor layer may include a first active layer **126** of the first thin film transistor **120**, a second active layer **146** of the second thin film transistor **140**, and an auxiliary layer **163**. The semiconductor layer may include polycrystalline silicon, monocrystalline silicon, oxide semiconductor, and the like.

A first gate insulating layer **170** is disposed on the semiconductor layer. The first gate insulating layer **170** covers the semiconductor layer. The first gate insulating layer **170** may function as a gate insulating film of the thin film transistor. The first gate insulating layer **170** may include silicon oxide, silicon nitride, silicon oxynitride, aluminum oxide, tantalum oxide, hafnium oxide, zirconium oxide, titanium oxide, or the like. These may be used alone or in combination with each other.

A first conductive layer is disposed on the first gate insulating layer **170**. The first conductive layer may include a first gate electrode **121** disposed on the first active layer **126** of the first thin film transistor **120**, a second gate electrode **141** disposed on the second active layer **146** of the second thin film transistor **140**, and a power wiring **161** disposed on the auxiliary layer **163**, with the first gate insulating layer **170** interposed therebetween, respectively. The first conductive layer may include at least one metal selected from the group consisting of molybdenum (Mo), aluminum (Al), platinum (Pt), palladium (Pd), silver (Ag), magnesium (Mg), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), calcium (Ca), titanium (Ti), tantalum (Ta), tungsten (W) and copper (Cu). The first conductive layer may be a single layer or a multilayer.

A second gate insulating layer **180** is disposed on the first conductive layer. The second gate insulating layer **180** may be an interlayer insulating layer. The second gate insulating layer **180** may be formed of an inorganic insulating material such as silicon oxide, silicon nitride, silicon oxynitride, hafnium oxide, aluminum oxide, titanium oxide, tantalum oxide, zinc oxide and the like.

A second conductive layer is disposed on the second gate insulating layer **180**. The second conductive layer includes a capacitor electrode **128** disposed on the first gate electrode

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121, with the second gate insulating layer 180 interposed therebetween. The capacitor electrode 128 may form a storage capacitor in cooperation with the first gate electrode 121.

In the same way as the first conductive layer described above, the second conductive layer may include at least one metal selected from the group consisting of molybdenum (Mo), aluminum (Al), platinum (Pt), palladium (Pd), silver (Ag), magnesium (Mg), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), calcium (Ca), titanium (Ti), tantalum (Ta), tungsten (W) and copper (Cu).

An interlayer insulating layer 190 is disposed on the second conductive layer. The interlayer insulating layer 190 may be an interlayer insulating film. Further, the interlayer insulating layer 190 may perform a surface planarization function. The interlayer insulating layer 190 may include an organic insulating material selected from the group consisting of acrylic resin, epoxy resin, phenolic resin, polyamide resin, polyimide resin, unsaturated polyester resin, polyphenylene resin, polyphenylenesulfide resin and benzocyclobutene (BCB).

A third conductive layer is disposed on the interlayer insulating layer 190. The third conductive layer includes a first drain electrode 123 and a first source electrode 124 of the first thin film transistor 120, a second drain electrode 143 and a second source electrode 144 of the second thin film transistor 140, and a power electrode 162 disposed on the power wiring 161.

The first source electrode 124 and the first drain electrode 123 may be electrically connected to the first active layer 126 through a first contact hole 129 passing through the interlayer insulating layer 190 and the second gate insulating layer 180. The second source electrode 144 and the second drain electrode 143 may be electrically connected to the second active layer 146 through a second contact hole 149 passing through the interlayer insulating layer 190 and the second gate insulating layer 180. The power electrode 162 may be electrically connected to the power wiring 161 through a third contact hole 169 passing through the interlayer insulating layer 190 and the second gate insulating layer 180.

The third conductive layer may include at least one metal selected from the group consisting of aluminum (Al), molybdenum (Mo), platinum (Pt), palladium (Pd), silver (Ag), magnesium (Mg), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), calcium (Ca), titanium (Ti), tantalum (Ta), tungsten (W) and copper (Cu). The third conductive layer may be a single layer or a multilayer. For example, the third conductive layer may have a stacked structure of Ti/Al/Ti, Mo/Al/Mo, Mo/AlGe/Mo, or Ti/Cu.

An insulating substrate layer 200 is disposed on the third conductive layer. The insulating substrate layer 200 may be formed of an organic insulating material selected from the group consisting of acrylic resin, epoxy resin, phenolic resin, polyamide resin, polyimide resin, unsaturated polyester resin, polyphenylene resin, polyphenylenesulfide resin and benzocyclobutene (BCB). The surface of the insulating substrate layer 200 may be flat.

The insulating substrate layer 200 may include a plurality of banks 410 and 420. The plurality of banks 410 and 420 may be arranged to face each other, keeping a distance therebetween, inside each pixel PX, and the distanced banks 410 and 420, e.g., a first bank 410 and a second bank 420, may have the first electrode 210 and the second electrode 220 arranged respectively thereon. As shown in FIG. 1, three banks 410 and 420, i.e., two first banks 410 and one second bank 420, are arranged to be covered by the respective first

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and second electrodes 210 and 220 within one pixel PX. Although FIG. 2 shows only a cross section of one first bank 410 and one second bank 420 among them, the arrangement configuration thereof may be identically applicable to the other first bank 410 not shown in FIG. 2.

However, the number of banks 410 and 420 is not limited thereto. For example, it may be possible that more banks 410 and 420 are arranged in one pixel PX along with more first and second electrodes 210 and 220. The banks 410 and 420 may include at least one first bank 410 on which the first electrode 210 is arranged and at least one second bank 420 on which the second electrode 220 is arranged. In this case, the first and second banks 410 and 420 may be arranged to face each other keeping a distance therebetween such that the plurality of banks are alternately arranged in one direction. In some embodiments, it may be possible that two first banks 410 are arranged, keeping a distance therebetween, and one second bank 420 is arranged between the distanced first banks 410.

Furthermore, although not shown in FIG. 2, the first and second electrodes 210 and 220 may include the respective electrode stems 210S and 220S and the respective electrode branches 210B and 220B as described above. It may be understood in FIG. 2 that the first and second electrode branches 210B and 220B are respectively arranged on the first and second banks 410 and 420.

The plurality of banks 410 and 420 may be formed with the substantially same material in a single process. In this case, the banks 410 and 420 may form a grid pattern. The banks 410 and 420 may include polyimide.

Meanwhile, although not shown in the drawing, at least part of the plurality of banks 410 and 420 may be arranged on a boundary of the pixels PX to make them distinct. In this case, the electrodes 210 and 220 may not be disposed on the banks 410 and 420 disposed at the boundary of the pixel PX. Such banks may be arranged in a substantially grid pattern along with the aforementioned first and second banks 410 and 420. At least part of the banks 410 and 420 arranged on the boundary of the pixels PX may be formed to cover electrode lines of the display device 10.

The plurality of banks 410 and 420 may each have a structure protruding at least partially from the insulating substrate layer 200. The banks 410 and 420 may protrude upward from a flat plane on which the light emitting elements 300 are arranged, such that a protruding part may at least partially have slopes. The banks 410 and 420 having a protruded structure with slopes may have reflection layers 211 and 221 arranged thereon to reflect incident light. The light directed from the light emitting element 300 to the reflection layers 211 and 221 may be reflected to the outside of the display device 10, i.e., upward from the banks 410 and 420. The banks 410 and 420 with the protruded structure may not be limited in shape. Although it is shown in FIG. 2 that the banks have a shape with a flat top surface and angular corners, the present disclosure is not limited thereto, and the banks may be protruded to have round corners.

The plurality of banks 410 and 420 may have reflection layers 211 and 221 arranged thereon.

The first reflection layer 211 covers the first bank 410 and is partially electrically connected to the first drain electrode 123 of the first thin film transistor 120 via a fourth contact hole 319_1 penetrating the insulating substrate layer 200. The second reflection layer 221 covers the second bank 420 and is partially electrically connected to the power electrode 162 via a fifth contact hole 319_2 penetrating the insulating substrate layer 200.

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The first reflection layer **211** may be electrically connected to the first drain electrode **123** of the first thin film transistor **120** via the fourth contact hole **319_1** within the pixel PX. Accordingly, the first thin film transistor **120** may be arranged in an area overlapping the pixel PX. FIG. 1 shows electrical connection to the first thin film transistor **120** via the first electrode contact hole CNTD arranged on the first electrode stem **210S**. That is, the first electrode contact hole CNTD may be the fourth contact hole **319_1**.

The second reflection layer **221** may also be electrically connected to the power electrode **162** via the fifth contact hole **319_2** within the pixel PX. FIG. 2 shows that the second reflection layer **221** is connected through the fifth contact hole **319_2** within one pixel PX. FIG. 1 shows that the second electrode **220** of each pixel PX is electrically connected to the power wiring **161** via the plurality of second electrode contact holes CNTS on the second electrode stem **220S**. That is, the second contact holes CNTS may be the fifth contact hole **319_2**.

As described with reference to FIG. 1, the first and second contact holes CNTD and CNTS may be respectively arranged on the first and second electrode stems **210S** and **220S**. In this respect, FIG. 2 shows that, in the cross-sectional view of the display device **10**, the first and second electrodes **210** and **220** are electrically connected to the first thin film transistor **120**, or the power wiring **161**, via the respective fourth and fifth contact holes **319_1** and **319_2**, in an area separated from the banks **410** and **420** on which the first and second electrode branches **210B** and **220B** are arranged.

However, the present disclosure is not limited thereto. For example, in FIG. 1, the second electrode contact holes CNTS may be arranged at various positions on the second electrode stem **220S** and, in some cases, on the second electrode branch **220B**. In some embodiments, the second reflection layer **221** may also be connected to one second electrode contact hole CNTS, or the fifth contact hole **319_2**, in an area out of one pixel PX.

In an area outside the emission area in which the pixels PX of the display device **100** are arranged, e.g., an outside area of the emission area, there may be a non-emission area in which no light emitting elements **300** are arranged. As described above, the second electrodes **220** of each pixel PX may be electrically connected via the second electrode stem **220S**, so as to receive the same electrical signal.

In some embodiments, in the case of the second electrode **220**, the second electrode stem **220S** may be electrically connected to the power electrode **162** via one second electrode contact hole CNTS in the non-emission area as the outside area of the display device **10**. Unlike the display device **10** of FIG. 1, because the second electrode stem **220S** is arranged to extend to adjacent pixels and be electrically connected to each other even though the second electrode stem **220S** is connected to the power electrode **162** via one contact hole, it may be possible to apply the same electrical signal to the second electrode branches **220B** of the respective pixels PX. In the case of the second electrode **220** of the display device **10**, the position of the contact hole for receiving an electrical signal from the power electrode **162** may vary according to the structure of the display device **10**.

Meanwhile, with reference back to FIGS. 1 and 2, the reflection layers **211** and **221** may include a material having high reflectivity for reflecting the light emitted from the light emitting elements **300**. For example, the reflection layers **211** and **221** may include, but are not limited to, a material such as silver (Ag) and copper (Cu).

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The first and second reflection layers **211** and **221** may include first and second electrode layers **212** and **222** arranged respectively thereon.

The first electrode layer **212** may be arranged directly on the first reflection layer **211**. The first electrode layer **212** may have a pattern substantially identical with that of the first reflection layer **211**. The second electrode layer **222** may be arranged directly on the second reflection layer **221** to be spaced apart from the first electrode layer **212**. The second electrode layer **222** may have a pattern that is substantially identical with that of the second reflection layer **221**.

In an embodiment, the electrode layers **212** and **222** may cover the reflection layers **211** and **221** respectively therebeneath. That is, the electrode layers **212** and **222** may be formed to be larger in size than the reflection layers **211** and **221** to cover the side end surfaces of the reflection layers **211** and **221**. However, the present disclosure is not limited thereto.

The first and second electrode layers **212** and **222** may transfer, to contact electrodes **261** and **262** (to be described later), an electrical signal directed to the first and second reflection layers **211** and **221** connected to the first thin film transistor **120** or the power electrode **162**. The electrode layers **212** and **222** may include a transparent conductive material. For example, the electrode layers **212** and **222** may include a material such as indium tin oxide (ITO), indium zinc oxide (IZO), and indium tin zinc oxide (ITZO), but are not limited thereto. In some embodiments, the reflective layers **211** and **221** and the electrode layers **212** and **222** may have a structure in which at least one transparent conductive layer such as ITO, IZO, or ITZO and at least one metal layer such as silver (Ag) or copper (Cu) are stacked. For example, the reflective layers **211** and **221** and the electrode layers **212** and **222** may have a stacked structure of ITO/Ag/ITO.

The first reflective layer **211** and the first electrode layer **212** disposed on the first bank **410** form the first electrode **210**. The first electrode **210** may protrude to regions extending from both ends of the first bank **410**, and accordingly, the first electrode **210** may contact the insulating substrate layer **200** in the protruding region. The second reflective layer **221** and the second electrode layer **222** disposed on the second bank **420** form the second electrode **220**. The second electrode **220** may protrude to regions extending from both ends of the second bank **420**, and accordingly, the second electrode **220** may contact the insulating substrate layer **200** in the protruding region.

The first and second electrodes **210** and **220** may be respectively arranged to cover the entire areas of the first and second banks **410** and **420**. However, as described above, the first and second electrodes **210** and **220** are arranged to face each other keeping a distance therebetween. Between the electrodes, a first insulating material layer **510**, which is to be described later, may be arranged, and the light emitting elements **300** may be arranged thereon.

In addition, the first reflective layer **211** may receive a driving voltage from the first thin film transistor **120** and the second reflective layer **221** may receive a source voltage from the power wiring **161**. Thus, the first electrode **210** and the second electrode **220** receive the driving voltage and the source voltage, respectively. The first electrode **210** may be electrically connected to the first thin film transistor **120**, and the second electrode **220** may be electrically connected to the power wiring **161**. Accordingly, the first and second contact electrodes **261** and **262** arranged respectively on the first and second electrodes **210** and **220** may receive the driving voltage and source voltage. The driving voltage and

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the source voltage may be transferred to the light emitting elements **300** such that the light emitting elements **300** emit light with a predetermined electric current flowing there-through.

The first insulating material layer **510** is arranged to partially cover the first and second electrodes **210** and **220**. The first insulating material layer **510** may be arranged to mostly cover the top surfaces of the first and second electrodes **210** and **220** and partially expose the first and second electrodes **210** and **220**. The first insulating material layer **510** may also be arranged in the space between the first and second electrodes **210** and **220**. The first insulating material layer **510** may have an islet or line shape formed along the space between the first and second electrode branches **210B** and **220B** in plan view.

FIG. 2 shows that the first insulating material layer **510** is arranged in the space between one first electrode **210** (e.g., first electrode branch **210B**) and one second electrode **220** (e.g., second electrode branch **220B**). However, as described above, there may be a plurality of the first and second electrodes **210** and **220**, such that the first insulating material layer **510** may be also arranged between one first electrode **210** and another second electrode **220** or between one second electrode **220** and another first electrode **210**. The first insulating material layer **510** may be arranged to partially cover the side surfaces of the first and second electrodes **210** and **220** that are opposite to the side surfaces facing each other. That is, the first insulating material layer **510** may be arranged to expose center parts of the first and second electrodes **210** and **220**.

On the first insulating material layer **510**, the light emitting element **300** is arranged. The first insulating material layer **510** may be arranged between the light emitting element **300** and the insulating substrate layer **200**. The first insulating material layer **510** may have a bottom surface contacting the insulating substrate layer **200**, and the light emitting element **300** may be arranged on the top surface of the first insulating material layer **510**. The first insulating material layer **510** may contact the electrodes **210** and **220** at both side surfaces thereof to electrically insulate the first and second electrodes **210** and **220** from each other.

The first insulating material layer **510** may overlap a partial area on the electrodes **210** and **220**, e.g., part of the area protruding in a direction in which the first and second electrodes **210** and **220** face each other. The first insulating material layer **510** may also be arranged on the areas where the sloping surfaces and flat top surfaces of the banks **410** and **420** overlap the electrodes **210** and **220**.

For example, the first insulating material layer **510** may cover the end parts protruding in the direction in which the first and second electrodes **210** and **220** face each other. The first insulating material layer **510** may contact the insulating substrate layer **200** partially on the bottom surface of the first insulating material layer **510**, and may contact the electrodes **210** and **220** partially on the bottom surface of the first insulating material layer **510** and on the side surfaces thereof. Accordingly, the first insulating material layer **510** may protect regions overlapping the respective electrodes **210** and **220** and electrically insulate them. Further, the first insulating material layer **510** may prevent a first conductivity type semiconductor **310** and a second conductivity type semiconductor **320** of the light emitting element **300** from directly contacting other members, thereby preventing damage to the light emitting element **300**.

However, the present disclosure is not limited thereto, and the first insulating material layer **510** may be arranged only on the areas overlapping the sloping side surfaces of the

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banks **410** and **420** in the areas on the first and second electrodes **210** and **220** in some embodiments. In this case, the bottom surface of the first insulating material layer **510** may terminate on the sloping side surfaces of the banks **410** and **420**, and the electrodes **210** and **220** arranged on part of the sloping side surfaces of the banks **410** and **420** may be exposed to contact the contact electrodes **260**.

The first insulating material layer **510** may also be arranged to expose both ends of the light emitting element **300**. Accordingly, the contact electrodes **260** may contact the exposed top surfaces of the electrodes **210** and **220** and both ends of the light emitting element **300**, and the contact electrode **260** may transfer the electrical signal applied to the first and second electrodes **210** and **220** to the light emitting element **300**.

At least one light emitting element **300** may be disposed between the first electrode **210** and the second electrode **220**. Although it is shown in FIG. 2 that one light emitting element **300** is arranged between the first and second electrodes **210** and **220**, it is apparent that a plurality light emitting elements **300** may be arranged in a different direction (e.g., second direction **D2**) in plan view as shown in FIG. 1.

In detail, the light emitting element **300** may be electrically connected to the first electrode **210** at one end thereof and the second electrode **220** at the other end thereof. The both ends of the light emitting elements **300** may respectively contact the first and second contact electrodes **261** and **262**.

Meanwhile, FIG. 1 exemplifies the case where only the light emitting elements **300** emitting the same color light are arranged in each pixel PX. However, the present disclosure is not limited thereto, and as described above, the light emitting elements **300** emitting light of different colors may be disposed together in one pixel PX.

The light emitting element **300** may be a light emitting diode. The light emitting element **300** may be a nanostructure mostly having a nano-size. The light emitting element **300** may be an inorganic light emitting diode made of an inorganic material. When the light emitting element **300** is an inorganic light emitting diode, a light emitting material having an inorganic crystal structure is disposed between two electrodes facing each other and an electric field is formed in a specific direction in the light emitting material. Then, the inorganic light emitting diode may be aligned between the two electrodes having a specific polarity.

In some embodiments, the light emitting element **300** may have a structure in which a first conductivity type semiconductor **310**, an element active layer **330**, a second conductivity type semiconductor **320**, and an electrode material layer **370** are sequentially formed. The light emitting element **300** may be manufactured by depositing, horizontally, the first conductivity type semiconductor **310**, the element active layer **330**, and the second conductivity type semiconductor **320**, and the electrode material layer **370** in the order on the insulating substrate layer **200**. That is, the light emitting elements **300** formed by depositing the plurality of layers may be arranged in the widthwise direction parallel with the insulating substrate layer **200**. However, the present disclosure is not limited thereto, and the light emitting elements **300** may be manufactured such that the layers are deposited in the reverse order between the first and second electrodes **210** and **220**.

In addition, the light emitting element **300** may include a plurality of insulating layers **380** surrounding the outer circumferential surfaces of the formed members. The insulating layer **380** may include a first insulating layer **381** and

a second insulating layer **382** disposed to surround the first insulating layer **381**. The insulating layer **380** may protect the formed members, and at the same time, any one insulating layer may function to protect another insulating layer. For example, when manufacturing the light emitting element **300**, the second insulating layer **382** may be disposed to surround the first insulating layer **381**, and may include a material having an etch selectivity different from that of the first insulating layer **381**, to protect the first insulating layer **381**. Accordingly, damage to the first insulating layer **381** that may occur when the light emitting element **300** is manufactured can be prevented. A more detailed description will be given later.

The second insulating material layer **520** may be arranged to overlap at least part of the light emitting element **300**. The second insulating material layer **520** may protect the light emitting element **300**, and simultaneously fix the light emitting element **300** between the first and second electrodes **210** and **220**.

Although it is shown in FIG. 2 that the second insulating material layer **520** is arranged only on the top surface of the light emitting element **300** in cross-sectional view, the second insulating material layer **520** may be arranged to surround the outer surface of the light emitting element **300**. That is, like the first insulating material layer **510**, the second insulating material layer **520** may be arranged to have an islet or line shape extending in the second direction D2 along the space between the first and second electrode branches **210B** and **220B** in plan view.

Part of the material of the second insulating material layer **520** may also be arranged at the area where the bottom surface of the light emitting element **300** and the first insulating material layer **510** overlap each other. That part may be formed when the light emitting element **300** is aligned on the first insulating layer **510**, and then the second insulating material layer **520** is disposed thereon during the manufacture of the display device **10**. That part may also be formed by the second insulating layer **520** partially permeating, during the formation of the second insulating layer **520**, into pores formed in a section of the first insulating material layer **510** contacting the bottom surface of the light emitting element **300**.

The second insulating material layer **520** may be arranged to expose both end surfaces of the light emitting element **300**. That is, in cross-sectional view, the second insulating material layer **520** arranged on the top surface of the light emitting element **300** is shorter in length, measured in an axis direction than the light emitting element **300**, such that the second insulating material layer **520** may be contracted inward in comparison with the both ends of the light emitting element **300**. Accordingly, the first insulating material layer **510**, the light emitting element **300**, and the second insulating material layer **520** may be deposited such that the side surfaces thereof are aligned in a stepwise manner. This may facilitate contact between the contact electrodes **261** and **262** and both end surfaces of the light emitting element **300**. However, the present disclosure is not limited thereto. The second insulating material layer **520** and the light emitting element **300** may have the same length, and both sides thereof may be aligned.

Meanwhile, the second insulating material layer **520** may be formed in a way of depositing the corresponding material on the first insulating material layer **510** and patterning the corresponding material in an area, e.g., area exposed for contact of the light emitting element **300** to the contact electrode **260**. Patterning the second insulating material layer **520** may be performed with a conventional dry etching

or wet etching process. Here, the first and second insulating material layers **510** and **520** may include materials different in etch selectivity to prevent the first insulating material layer **510** from being patterned. That is, the first insulating material layer **510** may serve as an etching stopper in patterning the second insulating material layer **520**.

Accordingly, the first insulating material layer **510** may not undergo material damage even when the second insulating material layer **520** covering the outer surface of the light emitting element **300** is patterned to expose the both ends of the light emitting element **300**. In particular, the first insulating material layer **510** and the light emitting element **300** may have smooth contact surfaces at the both ends of the light emitting element **300**, where the light emitting element **300** and the contact electrode **260** contact each other.

On the second insulating material layer **520**, the first contact electrode **261** disposed on the first electrode **210** and overlapping at least part of the second insulating material layer **520**, and the second contact electrode **262** disposed on the second electrode **220** and overlapping at least part of the second insulating material layer **520**, may be arranged.

The first and second contact electrodes **261** and **262** may be respectively arranged on the top surfaces of the first and second electrodes **210** and **220**. In detail, the first and second contact electrodes **261** and **262** may respectively contact the first and second electrode layers **212** and **222** in the area where the first insulating material layer **510** is patterned to expose parts of the first and second electrodes **210** and **220**. The first and second contact electrodes **261** and **262** may contact one end side of the light emitting element **300**, e.g., the first conductivity type semiconductor **310**, the second conductivity type semiconductor **320**, or the electrode material layer **370**. Accordingly, the first and second contact electrodes **261** and **262** may transfer the electrical signal applied to the first and second electrode layers **212** and **222** to the light emitting element **300**.

The first contact electrode **261** may be arranged on the first electrode **210** to cover the first electrode **210** in part and contact the light emitting element **300** and the first and second insulating material layers **510** and **520** in part, on the bottom surface of the first contact electrode **261**. One end of the first contact electrode **261** that is oriented to the second contact electrode **262** is arranged on the second insulating material layer **520**. The second contact electrode **262** may be arranged on the first electrode **210** to cover the second electrode **220** in part and contact the light emitting element **300**, the first insulating material layer **510**, and a third insulating material layer **530** in part on the bottom surface of the second contact electrode **262**. One end of the second contact electrode **262** that is oriented to the first contact electrode **261** is arranged on the third insulating material layer **530**.

The first and second insulating material layers **510** and **520** may be patterned into an area to cover the first and second electrodes **210** and **220** on the top surface of the first and second banks **410** and **420**. Accordingly, the first and second electrode layers **212** and **222** of the respective first and second electrodes **210** and **220** may be exposed to be electrically connected to the respective contact electrodes **261** and **262**.

The first contact electrode **261** and the second contact electrode **262** may be spaced apart from each other on the second insulating material layer **520** or the third insulating material layer **530**. That is, the first and second contact electrodes **261** and **262** may be arranged to contact the light emitting element **300** and the second insulating material

layer 520, or the third insulating material layers 530, together and, on the second insulating material layer 520, to be spaced apart in the deposition direction for electrical insulation. Accordingly, the first and second contact electrodes 261 and 262 may respectively receive different powers from the first thin film transistor 120 and the power wiring 161. For example, the first contact electrode 261 may receive a driving voltage applied from the first thin film transistor 120 to the first electrode 210, and the second contact electrode 262 may receive a common source voltage applied from the power wiring 161 to the second electrode 220. However, the present disclosure is not limited thereto.

The contact electrodes 261 and 262 may include a conductive material. For example, they may include ITO, IZO, ITZO, aluminum (Al), or the like. However, the present disclosure is not limited thereto.

Further, the contact electrodes 261 and 262 may include the same material as the electrode layers 212 and 222. The contact electrodes 261 and 262 may be arranged to have substantially the same pattern on the electrode layers 212 and 222, to contact the electrode layers 212 and 222. For example, the first and second contact electrodes 261 and 262 contacting the first and second electrode layers 212 and 222 may transfer the electrical signals applied to the first and second electrode layers 212 and 222 to the light emitting element 300.

The third insulating material layer 530 may be arranged on the first contact electrode 261 to electrically insulate the first and second contact electrodes 261 and 262 from each other. The third insulating material layer 530 may be arranged to cover the first contact electrode 261, and not to overlap an area of the light emitting element 300 such that the light emitting element 300 contacts the second contact electrode 262. The third insulating material layer 530 may partially contact the first contact electrode 261, the second contact electrode 262, and the second insulating material layer 520 on the top surface of the second insulating material layer 520. The third insulating material layer 530 may be disposed to cover one end of the first contact electrode 261 on the top surface of the second insulating material layer 520. Accordingly, the third insulating material layer 530 may protect the first contact electrode 261, and electrically insulate the first contact electrode 261 from the second contact electrode 262.

One end of the third insulating material layer 530 that is oriented to the second electrode 220 may be aligned with one side surface of the second insulating material layer 520.

Meanwhile, in some embodiments, the third insulating material layer 530 may be omitted in the display device 10. Accordingly, the first contact electrode 261 and the second contact electrode 262 may be disposed on substantially the same plane, and may be electrically insulated from each other by a passivation layer 550 to be described later.

The passivation layer 550 may be formed on the third insulating material layer 530 and the second contact electrode 262 to protect members disposed on the insulating substrate layer 200 against the external environment. When the first contact electrode 261 and the second contact electrode 262 are exposed, a problem of disconnection of the contact electrode material may occur due to electrode damage, so it is required to cover them with the passivation layer 550. That is, the passivation layer 550 may be disposed to cover the first electrode 210, the second electrode 220, the light emitting element 300, and the like. In addition, as described above, when the third insulating material layer 530 is omitted, the passivation layer 550 may be formed on the first contact electrode 261 and the second contact elec-

trode 262. In this case, the passivation layer 550 may electrically insulate the first contact electrode 261 and the second contact electrode 262 from each other.

Each of the above-described first insulating material layer 510, second insulating material layer 520, third insulating material layer 530, and passivation layer 550 may include an inorganic insulating material. For example, the first insulating material layer 510, the second insulating material layer 520, the third insulating material layer 530, and the passivation layer 550 may include a material such as silicon oxide (SiOx), silicon nitride (SiNx), silicon oxynitride (SiOxNy), aluminum oxide (Al₂O₃), aluminum nitride (AlN), and the like. The first insulating material layer 510, the second insulating material layer 520, the third insulating material layer 530, and the passivation layer 550 may be made of the same material, but may also be made of different materials. In addition, various materials that impart insulating properties to the first insulating material layer 510, the second insulating material layer 520, the third insulating material layer 530, and the passivation layer 550 are applicable.

Meanwhile, the first and second insulating material layers 510 and 520 may differ in etch selectivity as described above. As one example, when the first insulating material layer 510 includes silicon oxide (SiOx), the second insulating material layer 520 may include silicon nitride (SiNx). As another example, when the first insulating material layer 510 includes silicon nitride (SiNx), the second insulating material layer 520 may include silicon oxide (SiOx). However, the present disclosure is not limited thereto.

Meanwhile, the light emitting element 300 may be manufactured on a substrate by epitaxial growth. A seed crystal layer for forming a semiconductor layer may be formed on the substrate, and a desired semiconductor material may be deposited to grow. Hereinafter, the structure of the light emitting element 300 according to various embodiments will be described in detail with reference to FIG. 3.

FIG. 3 is a schematic view of a light emitting element according to an embodiment. FIG. 4 is a cross-sectional view taken along line 3b-3b' of FIG. 3.

Referring to FIG. 3, the light emitting element 300 may include a plurality of conductivity type semiconductors 310 and 320, an element active layer 330, an electrode material layer 370, and a plurality of insulating layers 380. The electrical signal received through the first and second electrodes 210 and 220 may be transferred to the element active layer 330 via the plurality of conductivity type semiconductors 310 and 320 to emit light.

In detail, the light emitting element 300 may include the first conductivity type semiconductor 310, the second conductivity type semiconductor 320, the element active layer 330 arranged between the first and second conductivity type semiconductors 310 and 320, the electrode material layer 370 arranged on the second conductivity type semiconductor 320, and a plurality of insulating layers 380 disposed to surround outer circumferential surfaces thereof. The plurality of insulating layers 380 may include the first insulating layer 381 in contact with the first conductivity type semiconductor 310, the second conductivity type semiconductor 320, the element active layer 330, and the electrode material layer 370 to surround outer circumferential surfaces thereof, and the second insulating layer 382 surrounding the first insulating layer 381. Although it is shown in FIG. 3 that the light emitting element 300 has a structure in which the first conductivity type semiconductor 310, the element active layer 330, the second conductivity type semiconductor 320, and the electrode material layer 370 are formed in order in the lengthwise direction thereof, the present disclosure is not

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limited thereto. The electrode material layer 370 may be omitted and, in some embodiments, it may be arranged on at least one of both side surfaces of each of the first and second conductivity type semiconductor 310 and 320. Hereinafter, a description is made of the exemplary light emitting element 300 of FIG. 3, and it is obvious that the following description of the light emitting element 300 is identically applicable to light emitting elements 300 including different structures.

The first conductivity type semiconductor 310 may be an n-type semiconductor layer. As one example, when the light emitting element 300 emits light of a blue wavelength band, the first conductivity type semiconductor 310 may include a semiconductor material having a chemical formula of $\text{In}_x\text{Al}_{1-x}\text{Ga}_{1-y}\text{N}$ ($0 \leq x \leq 1$, $0 \leq y \leq 1$, $0 \leq x+y \leq 1$). For example, it may be any one or more of n-type doped InAlGaN , GaN , AlGaIn , InGaIn , AlIn and InN . The first conductivity type semiconductor 310 may be doped with a first conductive dopant. For example, the first conductive dopant may be Si, Ge, Sn, or the like. The length of the first conductivity type semiconductor 310 may have a range of 1.5 μm to 5 μm , but is not limited thereto.

The second conductivity type semiconductor 320 may be a p-type semiconductor layer. As one example, when the light emitting element 300 emits light of a blue wavelength band, the second conductivity type semiconductor 320 may include a semiconductor material having a chemical formula of $\text{In}_x\text{Al}_y\text{Ga}_{1-x-y}\text{N}$ ($0 \leq x \leq 1$, $0 \leq x+y \leq 1$). For example, it may be any one or more of p-type doped InAlGaIn , GaIn , AlGaIn , InGaIn , AlIn and InN . The second conductivity type semiconductor 320 may be doped with a second conductive dopant. For example, the second conductive dopant may be Mg, Zn, Ca, Se, Ba, or the like. The length of the second conductivity type semiconductor 320 may have a range of 0.08 μm to 0.25 μm , but is not limited thereto.

The element active layer 330 is disposed between the first conductivity type semiconductor 310 and the second conductivity type semiconductor 320, and may include a material having a single or multiple quantum well structure. When the element active layer 330 includes a material having a multiple quantum well structure, a plurality of quantum layers and well layers may be stacked alternately. The element active layer 330 may emit light by coupling of electron-hole pairs according to an electric signal applied through the first conductivity type semiconductor 310 and the second conductivity type semiconductor 320. For example, when the element active layer 330 emits light of a blue wavelength band, it may include a material such as AlGaIn or AlInGaIn . In particular, when the element active layer 330 has a multiple quantum well structure, in which quantum layers and well layers may be stacked alternately, the quantum layer may include a material such as AlGaIn or AlInGaIn , and the well layer may include a material such as GaIn or AlGaIn . However, the present disclosure is not limited thereto, and the element active layer 330 may have a structure, in which semiconductor materials having large band gap energy and semiconductor materials having small band gap energy are alternately stacked, and may include other Group III to V semiconductor materials according to the wavelength band of the emitted light. The light emitted by the element active layer 330 is not limited to light of a blue wavelength band, but may also be light of a red or green wavelength band in some cases. The length of the element active layer 330 may have a range of 0.05 μm to 0.25 μm , but is not limited thereto.

The light emitted from the element active layer 330 may be projected through both side surfaces, as well as the outer

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surface of the light emitting element 300 in a longitudinal direction. The directionality of light emitted from the element active layer 330 is not limited to one direction.

The electrode material layer 370 may be an ohmic contact electrode. However, the present disclosure is not limited thereto, and the electrode material layer 370 may be a Schottky contact electrode. The electrode material layer 370 may include conductive metal. For example, the electrode material layer 370 may include at least one of aluminum (Al), titanium (Ti), indium (In), gold (Au), or silver (Ag). The electrode material layer 370 may include the same material or different materials. However, the present disclosure is not limited thereto.

The first insulating layer 381 may be formed on the outside of the first conductivity type semiconductor 310, the second conductivity type semiconductor 320, the element active layer 330, and the electrode material layer 370, and may function to protect them. For example, the first insulating material layer 381 may be formed to surround the side surfaces of the above-mentioned members, and may not be formed at both ends of the light emitting element 300 in the longitudinal direction, e.g., at both ends where the first conductivity type semiconductor 310 and the electrode material layer 370 are disposed. However, the present disclosure is not limited thereto.

The first insulating material layer 381 may include materials having insulating properties, for example, silicon oxide (SiO_x), silicon nitride (SiN_x), silicon oxynitride (SiO_xN_y), aluminum nitride (AlN), aluminum oxide (Al_2O_3), and the like. Accordingly, an electrical short circuit that may occur when the element active layer 330 directly contacts the first electrode 210 or the second electrode 220 can be prevented. Furthermore, the first insulating material layer 381 covers the element active layer 330 to protect the outer circumferential surface of the light emitting element 300, which may prevent degradation in light emission efficiency.

Although it is illustrated in the drawing that the first insulating layer 381 is formed to extend in the longitudinal direction to cover the first conductivity type semiconductor 310 to the electrode material layer 370, the present disclosure is not limited thereto. The first insulating material layer 381 may cover only the first conductivity type semiconductor 310, the element active layer 330, and the second conductivity type semiconductor 320, or only part of the outer surface of the electrode material layer 370, and expose part of the outer surface of the electrode material layer 370.

The thickness of the first insulating material layer 381 may have a range of 0.5 μm to 1.5 μm , but is not limited thereto.

The second insulating layer 382 may be disposed to surround the outer circumferential surface of the first insulating layer 381, and the second insulating layer 382 may have substantially the same shape as the first insulating layer 381. The second insulating layer 382 includes a material having an etch selectivity different from that of the first insulating layer 381, thereby preventing damage to the first insulating layer 381 that may occur in an etching or separation step in the manufacture of the light emitting element 300. Thus, it is possible to perform a function of protecting the first insulating layer 381. For example, when the etchant for separation includes fluorine (F), the etch selectivity of the second insulating layer 382 for the etchant for separation may be greater than the etch selectivity of the first insulating layer 381 for the etchant for separation.

The process of manufacturing the light emitting element 300 may include forming the first insulating layer 381 on the element grown on the substrate, and then separating the

element by a chemical lift off (CLO) method. Here, the first insulating layer **381** may be partially damaged by the etchant used for separating the element. In order to prevent this, the second insulating layer **382** may be formed to surround the outer circumferential surface of the first insulating layer **381**, to prevent the first insulating layer **381** from being damaged by the etchant for separation. Since the second insulating layer **382** may include a material having an etch selectivity different from that of the first insulating layer **381**, the second insulating layer **382** may not be etched by the etchant for separation, and the element grown on the substrate may be separated.

According to an embodiment, the second insulating material layer **382** may include materials having insulating properties and an etch selectivity different from that of the first insulating material layer **381**, for example, silicon oxide (SiOx), silicon nitride (SiNx), silicon oxynitride (SiOxNy), aluminum nitride (AlN), aluminum oxide (Al₂O₃), and the like.

When the first insulating layer **381** includes aluminum oxide (Al₂O₃), the second insulating layer **382** may include silicon nitride (SiNx). However, the present disclosure is not limited thereto. Accordingly, when manufacturing the light emitting element **300**, the second insulating layer **382** may prevent damage to the first insulating layer **381**. The first insulating layer **381** may have a substantially constant thickness in a long axis direction, crossing both ends of the light emitting element **300**.

The thickness of the second insulating material layer **382** may have a range of 0.5 μm to 1.5 μm, but is not limited thereto.

Further, in some embodiments, the second insulating material layer **382** may have an outer circumferential surface which is surface-treated. As described above, when the light emitting elements **300** are aligned between the electrodes **210** and **220**, the plurality of light emitting elements **300** may be applied in a dispersed state in a solution. Here, in order to keep the light emitting elements **300** in a dispersed state without aggregation with other light emitting elements **300** adjacent in the solution, the surface of the second insulating layer **382** may be treated in a hydrophobic or hydrophilic manner, such that it can maintain a mutually dispersed state in the solution. Accordingly, when the light emitting elements **300** are aligned, they may be aligned without aggregation, between the first electrode **210** and the second electrode **220**.

The light emitting element **300** may have a cylindrical shape. As shown in FIG. 4, the cross section taken by halving the light emitting element **300** in the lengthwise direction crossing the two ends of the light emitting element **300** may have a rectangular shape. However, the shape of the light emitting element **300** is not limited thereto, and may have various shapes such as a regular cube, a rectangular parallelepiped and a hexagonal prism. The light emitting element **300** may have a length *l* of 1 μm to 10 μm or 2 μm to 5 μm, and preferably about 4 μm. In addition, the diameter of the light emitting element **300** may have a range of 400 nm to 700 nm, and preferably may be about 500 nm.

Although the following description is made of the exemplary light emitting element **300** shown in FIG. 3 for convenience of explanation, the present disclosure may be identically applicable to the light emitting elements including more electrode material layers **370** or other structures.

Meanwhile, FIG. 5 is an enlarged view of part A of FIG. 2.

Referring to FIG. 5, in the light emitting element **300**, in cross-sectional view crossing both ends thereof, the first

conductivity type semiconductor **310**, the element active layer **330**, the second conductivity type semiconductor **320**, and the electrode material layer **370** may be formed in a horizontal direction with the insulating substrate layer **200**, and the first insulating layer **381** and the second insulating layer **382** may be formed in a direction perpendicular to the insulating substrate layer **200**. That is, the light emitting element **300** may be disposed on the insulating substrate layer **200** to be perpendicular to the direction in which the above-mentioned members are formed.

As described above, when manufacturing the light emitting element **300**, since the first insulating layer **381** is protected without being damaged by the second insulating layer **382**, the outer circumferential surface of the light emitting element **300** may form a smooth surface, and the element active layer **330** may be prevented from contacting other members by the first insulating layer **381** and the second insulating layer **382**. That is, the first insulating layer **381** may have a substantially constant thickness in a long axis direction crossing both ends of the light emitting element **300**.

In addition, when manufacturing the light emitting element **300**, it can be separated while preventing the outer surface of the light emitting element **300** from being damaged by the second insulating layer **382**, so that both end surfaces of the light emitting element **300** can form smooth surfaces and can have a relatively low roughness. Accordingly, the first conductivity type semiconductor **310** of the light emitting element **300** may be formed to have a smooth surface, which prevents an open circuit problem from occurring when contacting the first contact electrode **261**.

On the plane (indicated by line 4a-4a' in FIG. 5) where one end surface of the light emitting element **300** and the first contact electrode **261** are in contact, the end surface of the light emitting element **300** is smooth, thereby preventing a disconnection problem in which the electrode material of the first contact electrode **261** is cut off. For example, if the end surface of the light emitting element **300** is rough or protruded, or recessed to form a slope, this may degrade the thin film step coverage of the contact electrode material when the first contact electrode **261** and the light emitting element **300** are in contact, which leads to a partial cutoff of the electrode material. That is, the faulty contact between the light emitting element **300** and the first contact electrode **261** at the contact area (4a-4a' of FIG. 5) may block an electrical signal from reaching the light emitting element **300**, leading to a light emission error.

Meanwhile, if the end surface of the light emitting element **300** is smooth as shown in FIG. 5, this makes it possible to prevent a disconnection problem of the contact electrode material from occurring at the area (4a-4a' of FIG. 5) where the light emitting element **300** and the contact electrode **260** contact each other. This may be able to improve reliability of the light emitting element **300** of the display device **10**. According to an embodiment, the end surface of the light emitting element **300** may have a roughness value of 8 nm Ra to 12 nm Ra. However, the present disclosure is not limited thereto. Meanwhile, although not shown in the drawing, the above-described approach may be identically applicable to the second conductivity type semiconductor **320**, contacting the second contact electrode **262** or the side surface formed by the electrode material layer **370**.

The smooth end surface of the light emitting element **300** may be formed by a chemical lift off (CLO) method in which the light emitting element **300** chemically removes and separates the separating layer **1300** (see FIG. 8) when the

light emitting element **300** is manufactured. That is, the light emitting element **300** may be separated from the lower substrate layer without any external physical force, by letting the material grown on the end surface of the light emitting element **300** be cut off by removing the separating layer **1300** on which the light emitting element **300** was grown.

Here, when manufacturing the light emitting element **300** by the chemical lift off (CLO) method, the second insulating layer **382** may be formed so as not to damage the outer circumferential surface of the light emitting element **300**. The second insulating layer **382** may include a material that is not etched by the etchant for separation used when removing the separating layer **1300**. Accordingly, even if the light emitting element **300** is manufactured by the chemical lift off (CLO) method, the light emitting element **300** according to an embodiment can prevent damage to the outer circumferential surface thereof. At the same time, a smooth surface can be formed such that both end surfaces of the light emitting element **300** are flat, and a disconnection problem of the materials of the contact electrodes **261** and **262** as described above can be prevented.

A description is made hereinafter of the method for manufacturing the light emitting element **300** in detail with reference to FIGS. 6 to 16.

FIGS. 6 to 16 are schematic cross-sectional views schematically showing a method for manufacturing a light emitting element according to an embodiment.

First, with reference to FIG. 6, a lower substrate layer **1000** including a base substrate **1100** and a buffer material layer **1200** formed on the base substrate **1100** is prepared. As shown in FIG. 6, the lower substrate layer **1000** may have a layered structure formed by depositing the base substrate **1100** and the buffer material layer **1200** in order.

The base substrate **1100** may include a transparent substrate such as a sapphire (Al_2O_3) substrate and a glass substrate. However, the present disclosure is not limited thereto, and it may be formed of a conductive substrate material such as GaN, SiC, ZnO, Si, GaP and GaAs. The following description is directed to an exemplary case where the base substrate **1100** is a sapphire (Al_2O_3) substrate. Although not limited, the base substrate **1100** may have, for example, a thickness in the range of 400 μm to 1500 μm .

On the base substrate **1100**, a plurality of conductivity type semiconductor layers are formed. The plurality of conductivity type semiconductor layers grown by an epitaxial growth method may be grown by forming a seed crystal and depositing a crystal material thereon. Here, the conductivity type semiconductor layer may be formed using one of electron beam deposition, physical vapor deposition (PVD), chemical vapor deposition (CVD), plasma laser deposition (PLD), dual-type thermal evaporation, sputtering, and metal organic chemical vapor deposition (MOCVD), preferably, using the metal organic chemical vapor deposition (MOCVD). However, the present disclosure is not limited thereto.

Typically, a precursor material for forming the plurality of conductivity type semiconductor layers may be selected to form a target material in a typically selectable range without any limitation. For example, the precursor material may be a metal precursor including an alkyl group such as a methyl group or an ethyl group. Examples of the precursor material may include, but are not limited to, trimethylgallium $\text{Ga}(\text{CH}_3)_3$, trimethylaluminum $\text{Al}(\text{CH}_3)_3$, and triethyl phosphate $(\text{C}_2\text{H}_5)_3\text{PO}_4$. Hereinafter, with the omission of the description of the method and processing conditions for forming the plurality of conductivity type semiconductor

layers, a description is made of the processing order of the method for manufacturing the light emitting element **300** and the layered structure of the light emitting element **300** in detail.

A buffer material layer **1200** is formed on the base substrate **1100**. Although it is shown in the drawing that one buffer material layer **1200** is deposited, the present disclosure is not limited thereto, and a plurality of layers may be formed.

At a step to be described later, a separating layer **1300** may be disposed on the buffer material layer **1200** and then a crystal for the first conductivity type semiconductor layer **3100** may grow on the separating layer **1300**. The buffer material layer **1200** may be interposed between the base substrate **1100** and the separating layer **1300** to reduce a grid constant difference of the first conductivity type semiconductor layer **3100**. Although the first conductivity type semiconductor layer **3100** may be directly formed on the separating layer **1300** disposed on the base substrate **1100**, the buffer material layer **1200** may provide the seed crystal to facilitate crystal growth to the first conductivity type semiconductor layer **3100**.

For example, the buffer material layer **1200** may include an undoped semiconductor, and may be a material including substantially the same material as the first conductivity type semiconductor layer **3100** that is neither n-type doped nor p-type doped. In an exemplary embodiment, the buffer material layer **1200** may be, but is not limited to, at least one of undoped InAlGa, GaN, AlGa, InGa, AlN, or InN.

Meanwhile, in some embodiments, a plurality of layers may be formed on the buffer material layer **1200**, and the separating layer **1300** may be deposited thereon. The buffer material layer **1200** may also be omitted depending on the base substrate **1100**. A detailed description thereof will be given with reference to other embodiments. Hereinafter, a description is made of the exemplary case where the buffer material layer **1200** including an undoped semiconductor material is formed on the base substrate **1100**.

Next, with reference to FIG. 7, the separating layer **1300** is formed on the lower substrate layer **1000**.

The separating layer **1300** may have the first conductivity type semiconductor layer **3100** formed thereon. That is, the separating layer **1300** may be interposed between the first conductivity type semiconductor layer **3100** and the buffer material layer **1200**, and the separating layer **1300** may include a material facilitating growth of the crystal of the first conductivity type semiconductor layer **3100**. The separating layer **1300** may include at least one of an insulating material or a conductive material. As one example, the separating layer **1300** may include silicon oxide (SiO_x), silicon nitride (SiN_x), silicon oxynitride (SiO_xNy) or the like as an insulating material, and may include ITO, IZO, IGO, ZnO, graphene, graphene oxide or the like as a conductive material. However, the present disclosure is not limited thereto.

The separating layer **1300** may be etched and removed at a step to be described later, thereby performing a function of separating the light emitting element **300** from the lower substrate layer **1000**. Removing the separating layer **1300** may be performed by the chemical lift off (CLO) method as described above, so that the end surface of the light emitting element **300** may be formed substantially the same as the surface of the separating layer **1300**. That is, the end surface of the light emitting element **300** may have a flat surface.

The separating layer **1300** may also serve as an etching stopper between a semiconductor structure **3000** and the buffer material layer **1200** during the process of etching the

semiconductor structure **3000**. That is, when the semiconductor structure **3000** is etched, the separating layer **1300** may be patterned simultaneously in one process, or patterned separately in a different process. There is no limitation on the method of manufacturing the light emitting element **300**.

However, the present disclosure is not limited thereto, and more separating layers **1300** may be arranged in the semiconductor structure **3000** or the lower substrate layer **1000**, and regions such as on the interface between the buffer material layer **1200** and the first conductivity type semiconductor layer **1300**.

Next, with reference to FIG. 8, the semiconductor structure **3000** is formed by forming a first conductivity type semiconductor layer **3100**, an active material layer **3300**, a second conductivity type semiconductor layer **3200**, and a conductive electrode material layer **3700** in order on the separating layer **1300**.

The semiconductor structure **3000** may be partially etched at a step to be described later to form an element rod (ROD). The plurality of material layers included in the semiconductor structure **3000** may be formed through a conventional process as described above. On the separating layer **1300**, the first conductivity type semiconductor layer **3100**, the active material layer **3300**, the second conductivity type semiconductor layer **3200**, and the conductive electrode material layer **3700** may be deposited in order, and they may respectively include the same materials as those of the first conductivity type semiconductor **310**, the element active layer **330**, the second conductivity type semiconductor **320**, and the electrode material layer **370** of the light emitting element **300**.

Meanwhile, the light emitting element **300** may be manufactured with the omission of the electrode material layer **370**, or with the further inclusion of a different electrode material layer **370** formed on the bottom surface of the first conductivity type semiconductor **310**. In other words, unlike FIG. 8, in the semiconductor structure **3000**, the conductive electrode material layer **3700** may be omitted, or another conductive electrode material layer may be formed below the first conductivity type semiconductor layer **3100**. The following description is made of the exemplary case where the semiconductor structure **3000** includes the conductive electrode material layer **3700**.

Next, referring to FIGS. 9 to 11, the element rod (ROD) is formed by etching the first conductivity type semiconductor layer **3100**, the active material layer **3300**, the second conductivity type semiconductor layer **3200**, and the electrode material layer **3700** in a direction perpendicular to the lower substrate layer **1000**.

First, with reference to FIGS. 9 and 10, forming the element rod ROD by vertically etching the semiconductor structure **3000** may include a patterning process that may be conventionally carried out. For example, forming the element rod ROD by etching the semiconductor structure **3000** may include forming an etching mask layer **1600** and an etching pattern layer **1700** on the semiconductor structure **3000**, and etching the semiconductor structure **3000** according to a pattern of the etching pattern layer **1700**, and removing the etching mask layer **1600** and the etching pattern layer **1700**.

The etching mask layer **1600** may serve as a mask for consecutively etching the first conductivity type semiconductor layer **3100**, the active material layer **3300**, the second conductivity type semiconductor layer **3200**, and the conductive electrode material layer **3700** of the semiconductor structure **3000**. The etching mask layer **1600** may include a

first etching mask layer **1610** including an insulating material and a second etching mask layer **1620** including metal.

The insulating material included in the first etching mask layer **1610** of the etching mask layer **1600** may be an oxide or a nitride. Examples of the insulating material may include silicon oxide (SiOx), silicon nitride (SiNx), and silicon oxynitride (SiOxNy). The first etching mask layer **1610** may have a thickness in the range of 0.5 μm to 1.5 μm without being limited thereto.

The second etching mask layer **1620** may not be limited in material as long as it can serve as a mask for consecutively etching the semiconductor structure **3000**. For example, the second etching mask layer **1620** may include chrome (Cr). The second etching mask **1620** may have a thickness in the range from 30 nm to 150 nm without being limited thereto.

The etching pattern layer **1700** formed on the etching mask layer **1600** may include at least one nanopattern separated from each other thereon. The etching pattern layer **1700** may serve as a mask for consecutively etching the semiconductor structure **3000**. There is no limitation on the etching method as long as it can form a pattern including a polymer, a polyethylene sphere, or a silica sphere on the etching pattern layer **1700**.

For example, in the case where the etching pattern layer **1700** includes a polymer, it may be possible to employ a conventional method for forming a pattern with the polymer. For example, it may be possible to use a method such as photolithography, e-beam lithography, nanoimprint lithography to form the etching pattern layer **1700** including the polymer.

Particularly, the structure, shape, and separation interval of the etching pattern layer **1700** may be associated with the shape of the finally manufactured light emitting element **300**. However, because the light emitting element **300** may have a different shape as described above, the etching pattern layer **1700** is not particularly limited in structure. For example, if the etching pattern layer **1700** has a pattern of circles separated from each other, the semiconductor structure **3000** may be vertically etched to manufacture the light emitting element **300** having a cylinder shape. However, the present disclosure is not limited thereto.

Next, the semiconductor structure **3000** may be etched according to the pattern of the etching pattern layer **1700** to form the element rod ROD. An area in which a plurality of nanopatterns are spaced apart in the etching pattern layer **1700** may be vertically etched to form a hole h. The hole h may be selectively formed from the etching mask layer **1600** to the area where the separating layer **1300** is formed.

The hole h may be formed using a conventional method. For example, the etching process may be performed with dry etching, wet etching, reactive ion etching (RIE), inductively coupled plasma reactive ion etching (ICP-RIE), or the like. The dry etching is capable of anisotropic etching, which may be appropriate for forming a hole h through vertical etching. In the case of using the aforementioned etching technique, it may be possible to use Cl_2 or O_2 as an etchant. However, the present disclosure is not limited thereto.

In some embodiments, etching the semiconductor structure **3000** may be carried out with a combination of the dry etching and the wet etching. For example, it may be possible to perform etching in a depth direction with the dry etching, and then anisotropic etching with the wet etching, such that the etched sidewalls are placed on the plane perpendicular to the surface.

Meanwhile, forming the element rod ROD by etching the semiconductor structure **3000** may include patterning the separating layer **1300** together during one etching process or

patterning part of the separating layer **1300** after the element rod ROD is formed through another etching process.

That is, the separating layer **1300** may be patterned together in the etching process of forming the hole by etching the semiconductor structure **3000** or patterned in a separate process after acting as an etching stopper in the process of etching the semiconductor structure **3000**.

For example, as shown in FIG. **10**, when patterning the semiconductor structure **3000**, when the etchant does not include an etchant for removing the separating layer **1300**, only the semiconductor structure **3000** is etched to form the hole **h**, and the separating layer **1300** is not etched and may function as an etching stopper. Accordingly, the element rod ROD may be formed in the state where the separating layer **1300** is not etched, and the separating layer **1300** may be patterned through a different etching process. On the other hand, although not shown in the drawing, when the etchant includes an etchant for removing the separating layer **1300**, the semiconductor structure **3000** and the separating layer **1300** may be etched together, but the present disclosure is not limited thereto.

As described with reference to FIG. **11**, the mask layer **1600** and the etching pattern layer **1700** remaining on the vertically etched semiconductor structure **3000** may be removed by a conventional method, e.g., dry etching and wet etching, to form the element rod ROD.

Next, referring to FIGS. **12** and **13**, a first insulating layer **3810** partially surrounding the outer surface of the element rod ROD is formed.

The first insulating material layer **3810** is an insulating material formed on the outer surface of the element rod ROD, and may be formed by depositing the insulating material on the outer surface of the vertically etched element rod ROD or dipping the element rod ROD in the insulating material, without being limited thereto. For example, the first insulating material layer **3810** may be formed using atomic layer deposition (ALD). The first insulating layer **3810** may form the first insulating layer **381** of the light emitting element **300**. As described above, the first insulating material layer **3810** may include a material such as silicon oxide (SiOx), silicon nitride (SiNx), silicon oxynitride (SiOxNy), aluminum oxide (Al₂O₃), and aluminum nitride (AlN).

Referring to FIG. **12**, the first insulating layer **3810** may be formed on the side and top surfaces of the element rod ROD, and on the separating layer **1300** or the buffer material layer **1200** exposed to the outside by etching the element rod ROD in a separated state. In order to expose both end side surfaces of the element rod ROD, the first insulating material layer **3810** formed on the top surface of the element rod ROD needs to be removed. Accordingly, it may be necessary to partially remove the first insulating material layer **3810** formed in a direction perpendicular to the lengthwise direction of the element rod ROD, i.e., the direction parallel with the base substrate **1100**. That is, as illustrated in FIG. **13**, at least the top surface of the element rod ROD and the first insulating layer **3810** disposed on the separating layer **1300** or the buffer material layer **1200** may be removed to expose the top surface of the element rod ROD. In order to accomplish this, a process such as dry etching as anisotropic etching or etch-back may be performed.

Next, referring to FIGS. **14** and **15**, a second insulating layer **3820** surrounding the outer circumferential surface of the first insulating layer **3810** is formed. The second insulating layer **3820** may have a different material from the first insulating layer **3810**, and they may be formed by substantially the same method.

The second insulating layer **3820** may form the second insulating layer **382** in the light emitting element **300** manufactured by separating the element rod ROD from the lower substrate layer **1000**. Separating the element rod ROD from the lower substrate layer **1000** may be performed by etching and removing the separating layer **1300**. In this case, the second insulating layer **3820** may prevent the first insulating layer **3810** from being damaged when the separating layer **1300** is etched.

According to an embodiment, the second insulating layer **3820** may include a material different in etch selectivity from the first insulating layer **3810** and the separating layer **1300**. The material that may be included in the second insulating layer **3820** may be substantially the same as the material that may be included in the second insulating layer **382** described above.

Finally, as shown in FIG. **16**, the separating layer **1300** on the lower substrate layer **1000** is removed to separate the element rod ROD, thereby manufacturing the light emitting element **300**.

Separating the element rod ROD may include removing the separating layer **1300** by the chemical lift off (CLO) method. In order to remove the separating layer **1300**, a wet etching process may be performed using an etchant for separation such as hydrofluoric acid (HF) or buffered oxide etch (BOE), but is not limited thereto.

As described above, the second insulating layer **3820** may be maintained without damaging the material by the etchant for separation, and may function to protect the first insulating layer **3810**. Accordingly, the outer circumferential surfaces of the manufactured light emitting element **300** may be maintained such that the first insulating layer **381** and the second insulating layer **382** are smooth. In addition, the first insulating layer **381** may have substantially uniform thickness in the long axis direction crossing both ends of the light emitting element **300**.

For example, when the separating layer **1300** is removed by the etchant for separation containing fluorine (F), the first insulating layer **3810** and the second insulating layer **3820** may not be damaged. That is, the second insulating layer **3820** may have an etch selectivity, with respect to the etchant for separation, that is greater than an etch selectivity of the first insulating layer **3810**, with respect to the etchant for separation. In addition, the second insulating layer **3820** may have an etch selectivity, with respect to the etchant for separation, that is greater than an etch selectivity of the separating layer **1300**, with respect to the etchant for separation.

For example, when the first insulating layer **3810** includes aluminum oxide (Al₂O₃) and the separating layer **1300** includes silicon oxide (SiOx), in the step of removing the separating layer **1300**, a part of the first insulating layer **3810** may be removed together by the etchant for separation. In this case, some members of the light emitting element **300** may be exposed to cause defects in the light emitting element **300**. In order to prevent this, when the second insulating layer **3820** including a material different in etch selectivity, for example, silicon nitride (SiNx), is formed on the outer circumferential surface of the first insulating layer **3810** to remove the separating layer **1300**, the first insulating layer **3810** may be protected from the etchant for separation.

Accordingly, even if the separating layer **1300** is removed, the manufactured light emitting element **300** may be formed such that the second insulating layer **382** surrounds the outer circumferential surface thereof, and may maintain a smooth surface without being damaged by the etchant for separation.

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Meanwhile, when the second insulating layer **3820** is omitted, a part of the material included in the first insulating layer **3810** may be damaged in the step of removing the separating layer **1300**. FIG. 17 is a schematic view showing a part of a method of manufacturing a light emitting element according to a comparative example.

Referring to FIG. 17, it can be seen that when the separating layer **1300** is removed without forming the second insulating layer **3820**, a part of the material included in the first insulating layer **3810** is damaged, and a first insulating layer **381'** of a manufactured light emitting element **300'** has a rough surface. In this case, a part of the element active layer **330** or the conductivity type semiconductors **310** and **320** of the light emitting element **300'** may be exposed and damaged, so that the light emitting element **300'** may be defective, and the first insulating layer **381'** having a rough surface may cause a contact failure between the light emitting element **300'** and the contact electrode **260**.

On the other hand, as illustrated in FIG. 16, after the second insulating layer **3820** is formed, the light emitting element **300** manufactured by removing the separating layer **1300** includes the second insulating layer **382**. Thus, it is possible to prevent damage to the materials of the first insulating layer **381** and the second insulating layer **382** by the etchant for separation, and to keep smooth the outer circumferential surface of the light emitting element **300**. Accordingly, the conductivity type semiconductors **310** and **320** or the element active layer **330** may be protected from exposure, and contact defects between the light emitting element **300** and the contact electrode **260** may be reduced.

In addition, the light emitting element **300** manufactured by the chemical lift off (CLO) method can maintain a flat and smooth end surface, and at the same time, the plurality of light emitting elements **300** can have uniformity of the end surfaces.

As described above, the method of manufacturing the light emitting element **300** according to an embodiment may include forming the second insulating layer **3820** to surround the first insulating layer **3810**, and removing the separating layer **1300** by the chemical lift off (CLO) method. The second insulating layer **3820** includes a material different in etch selectivity from the first insulating layer **3810** and the separating layer **1300**, so that when the separating layer **1300** is removed, damage to materials of the first insulating layer **3810** and the second insulating layer **3820** can be prevented. Accordingly, the manufactured light emitting element **300** can maintain a smooth surface without damaging the material of the outer circumferential surface thereof, and is separated by the chemical lift off (CLO) method, so that the end surface is flat and uniformity can be secured. In addition, the disconnection of the contact electrode material is prevented at both end surfaces of the light emitting element **300** contacting the contact electrode **260**, thereby improving light emission reliability of the display device **10**.

Meanwhile, the step of forming the first insulating layer **3810** and the second insulating layer **3820** includes a partial etching step to expose the top surface of the element rod ROD, respectively. Here, the etching step to expose the top surface of the element rod ROD may be performed by forming the first insulating layer **3810** and the second insulating layer **3820** and etching them simultaneously.

FIG. 18 is a schematic view showing a part of a method of manufacturing a light emitting element according to another embodiment.

Referring to FIG. 18, a first insulating layer **3810_1** may be formed to surround the outside of the element rod ROD,

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and a second insulating layer **3820_1** surrounding the outside of the first insulating layer **3810_1** may be formed before exposing the top surface of the element rod ROD. In other words, the step of exposing the top surface of the element rod ROD by etching a part of the first insulating layer **3810_1** is omitted, and the second insulating layer **3820_1** is formed directly on the first insulating layer **3810_1**. Thereafter, in order to expose the top surface of the element rod ROD, an etching process of partially removing the first insulating layer **3810_1** and the second insulating layer **3820_1** may be performed simultaneously.

In this case, since the process of forming the first insulating layer **3810_1** or the second insulating layer **3820_1** and the etching process for exposing the top surface of the element rod ROD can be performed at one time, there are advantages in the process. Also, in some cases, the first insulating layer **3810_1** and the second insulating layer **3820_1** may be removed through one etching process through dry etching.

Meanwhile, the arrangement of the separating layer **1300** is not limited to the arrangement shown in FIG. 7. The separating layer **1300** may have a pattern such that a portion of the buffer material layer **1200** of the lower substrate layer **1000** is exposed, and in some cases, a first sub-conductivity type semiconductor layer may be disposed on the buffer material layer **1200**, and the separating layer **1300** may be disposed thereon. Hereinafter, other embodiments of the separating layer **1300** will be described.

FIGS. 19 and 20 are cross-sectional views schematically showing the arrangement of a separating layer in a semiconductor structure according to another embodiment. According to an embodiment, a separating layer **1300_2** may have a plurality of separating layer masks **1310** spaced apart from each other to form a pattern.

According to FIG. 19, a first conductivity type semiconductor layer **3100_2** may be grown from a buffer material layer **1200_2**. In this case, defects between the grain boundaries of the first conductive type semiconductor layer **3100_2** may be reduced further than when the first conductive type semiconductor layer **3100_2** directly grows on the separating layer **1300_2**. The crystals of the first conductivity type semiconductor layer **3100_2** are grown between the separation spaces of the separating layer masks **1310** of the separating layer **1300_2**, and the crystals merge on the separating layer masks **1310**. Thus, a grain boundary may be formed only in a region where the crystals merge. That is, the number of defects between the grain boundaries in the first conductivity type semiconductor layer **3100_2** finally formed by forming the separating layer masks **1310** may be reduced.

Also, the separating layer **1300** may be disposed in the first conductivity type semiconductor layer **3100** of the semiconductor structure **3000**.

With reference to FIG. 20, a separating layer **1300_3** may be arranged on a first sub-conductivity type semiconductor layer **3100'_3** deposited on the buffer material layer **1200**, and a first conductivity type semiconductor layer **3100_3** may be deposited thereon. The first sub-conductivity type semiconductor layer **3100'_3** may include a material substantially identical with that of the first conductivity type semiconductor layer **3100_3**. That is, the separating layer **1300_3** may be arranged in the first conductivity type semiconductor layer **3100_3**.

As described above, the buffer material layer **1200** may provide a seed crystal of the first conductivity type semiconductor layer **3100** growing on the separating layer **1300**, and may reduce the grid constant of the interfaces. The

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semiconductor structure **3000** of FIG. **20** may facilitate crystal growth of the first conductivity type semiconductor layer **3100_3** by substantially including the separating layer **1300_3** in the first conductivity type semiconductor layer **3100_3**.

Meanwhile, the second insulating layer **382** of the light emitting element **300** may include a material having the same etch selectivity as the plurality of insulating layers of the display device **10**, for example, the second insulating material layer **520**. In this case, when the display device **10** is manufactured, a part of the second insulating layer **382** may be removed in a patterning step performed to contact the contact electrodes **260** at both ends of the light emitting element **300**. Accordingly, the first insulating layer **381** of the light emitting element **300** may be partially exposed, and the contact electrode **260** may partially contact the first insulating layer **381**.

FIG. **21** is a cross-sectional view illustrating a portion of a display device according to another embodiment.

According to an embodiment, the display device **10** may be disposed such that at least a portion of a second insulating layer **382_4** of a light emitting element **300_4** is removed, and the contact electrode **260** partially overlaps the first insulating layer **381_4**.

In the display device **10** shown in FIG. **21**, the second insulating layer **382_4** on the top surface of a light emitting element **300_4** in cross-sectional view is partially removed, and the first insulating layer **381_4** is partially exposed. Although only a first contact electrode **261_4** in contact with the first insulating layer **381_4** on one end side of the light emitting element **300_4** is shown in the drawing, it is obvious that the first insulating layer **381_4** on the other end side of the light emitting element **300_4** may contact the second contact electrode **262** in the same way.

Referring to FIG. **21**, the contact electrode **260** according to an embodiment may contact the exposed first insulating layer **381_4**. Unlike the display device **10** of FIG. **2**, when the second insulating layer **382_4** includes a material having the same etch selectivity as the second insulating material layer **520**, the second insulating layer **382_4** may be partially etched in the step of patterning the second insulating material layer **520** when manufacturing the display device **10**. Accordingly, since the first insulating layer **381_4** is partially exposed, and the first insulating layer **381_4** includes a material different in etch selectivity from the second insulating layer **382_4**, the first insulating layer **381_4** is hardly damaged. That is, the first conductivity type semiconductor **310**, the element active layer **330**, the second conductivity type semiconductor **320**, and the electrode material layer **370** of the light emitting element **300_4** can be protected by the first insulating layer **381_4**.

In concluding the detailed description, those skilled in the art will appreciate that many variations and modifications can be made to the preferred embodiments without substantially departing from the principles of the present invention. Therefore, the disclosed preferred embodiments of the invention are used in a generic and descriptive sense only and not for purposes of limitation.

The invention claimed is:

1. A manufacturing method of a light emitting element, the method comprising:

preparing a lower substrate comprising a substrate and a buffer semiconductor layer formed on a surface of the substrate;

forming an element rod by forming a separating layer on the lower substrate; forming a first conductivity type semiconductor layer, an active material layer, and a

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second conductivity type semiconductor layer on the separating layer; and etching the first conductivity type semiconductor layer, the active material layer, the second conductivity type semiconductor layer, and the separating layer in a first direction perpendicular to an extension direction of the surface of the substrate;

forming a first insulating layer surrounding an outer surface of the element rod along a second direction perpendicular to the first direction;

forming a second insulating layer surrounding an outer surface of the first insulating layer along the second direction; and

separating the element rod from the lower substrate to form a light emitting element,

wherein the forming of the first insulating layer comprises:

forming a first insulating layer to cover an outer surface and a top surface of the element rod; and

a first etching step of exposing the top surface of the element rod by etching the first insulating layer, and wherein the forming of the second insulating layer comprises:

forming a second insulating layer to cover the outer surface and the top surface of the element rod; and a second etching step of exposing the top surface of the element rod by etching the second insulating layer, and

wherein in the light emitting element, a parting surface where the element rod is separated by removing the separating layer has a surface roughness in a range of 8 nm Ra to 12 nm Ra.

2. The manufacturing method of claim 1, wherein the forming of the light emitting element comprises etching and removing the separating layer by an etchant for separation containing fluorine (F), and

the second insulating layer has an etch selectivity with respect to the etchant, which is greater than an etch selectivity of the separating layer with respect to the etchant.

3. The manufacturing method of claim 2, wherein the second insulating layer has an etch selectivity with respect to the etchant, which is greater than an etch selectivity of the first insulating layer with respect to the etchant.

4. The manufacturing method of claim 3, wherein the first insulating layer includes at least one of silicon oxide (SiO_x), aluminum oxide (Al_2O_3), and silicon oxynitride (SiO_xN_y), and the second insulating layer includes at least one of silicon nitride (SiN_x), aluminum nitride (AlN), and silicon oxynitride (SiO_xN_y).

5. The manufacturing method of claim 4, wherein in the light emitting element, the parting surface is flat and parallel to a top surface of the second conductivity type semiconductor layer.

6. The manufacturing method of claim 1, wherein the first insulating layer has a constant thickness in a long axis direction crossing both ends of the light emitting element.

7. The manufacturing method of claim 1, wherein in the first etching step and the second etching step, at least a part of the separating layer in an area overlapping a separation region of the element rod is exposed.

8. The manufacturing method of claim 7, wherein the second insulating layer is formed to surround an outer surface of the first insulating layer after forming the first insulating layer, and the first etching step and the second etching step are simultaneously performed after forming the second insulating layer.

9. The manufacturing method of claim 1, wherein the forming of the element rod further comprises forming an electrode material layer on the second conductivity type semiconductor layer.

10. The manufacturing method of claim 9, wherein the forming of the element rod further comprises:

forming an etching mask layer on the electrode material layer and an etching pattern layer having one or more nanopatterns separated from each other on the etching mask layer;

forming a hole by etching an area formed by the nanopatterns being separated from each other in a direction perpendicular to the lower substrate; and

removing the etching mask layer and the etching pattern layer.

11. The manufacturing method of claim 10, wherein the first conductivity type semiconductor layer, the active material layer, the second conductivity type semiconductor layer, and the electrode material layer include a material different in etch selectivity from the separating layer, and

the forming of the hole further comprises etching the first conductivity type semiconductor layer, the active material layer, the second conductivity type semiconductor layer, and the electrode material layer in a direction perpendicular to the lower substrate to expose at least a portion of an overlapping area between the separating layer and the area formed by the nanopatterns being separated from each other; and etching and patterning an exposed area of the separating layer.

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